ABSTRACT

The main objective of this thesis is to make a research on the importance of optimum design of electronic components. The life expectancy of typical electronic component depends on the operating voltage, current and temperature. Decreasing the temperature of electronic component at its operation can increase the life expectancy of electronic components. In thesis, the various approaches for cooling electronic circuit boards are discussed. Since the transformation of gasoline vehicle to electric vehicles is growing rapidly in this decade, the need for designing reliable circuits is an inevitable factor. One of the main component of electric automobiles is its motor controller board which is typically H-bridge circuit.

In this thesis a simplified version of motor controller board is selected for analysing the thermal efficiency. The geometry of circuit was fabricated in Ansys Icepack which is a widely used software package for analysing the thermal efficiency of components, with reference to the standard package sizes of electronic components available in market. After the preparation of geometry, quadratic order meshing of the geometry was carried out in Ansys Icepack. The study was executed in 4 stages. In the first stage thermal analysis of board was carried out considering as natural convection condition and results were analysed to find out the most heating element in circuit. Based on the results obtained from first stage in the second stage, heat sink was employed on the top of component which heats up largely. In the third stage optimization procedure was carried out to find the optimum design for heat sinks. In the fourth stage, the simulation environment is considered as forced convection and a model fan was installed in the board cabinet with its flow direction parallel to the direction of heatsink fins. Finally, interpretation of the results obtained in 4 stages is carried out.

Keywords: Thermal Optimisation, Ansys Icepak, H-bridge.

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1. INTRODUCTION

Electronic cooling is the process of controlling the temperature of components within an electronic system. Electronics' recent growth, along with their further shrinking, has resulted in a considerable boost in power density. As the move toward smaller circuits leads to more densely populated packaging paired with higher processing speeds, engineers are finding it more challenging than ever to design effective heat management systems. The ongoing downsizing of semiconductor and other mini- and micro-scale electronic technologies has resulted in a very significant rise in power density for high-performance processors. The two most significant problems are sufficient heat flux removal and extremely non-uniform power dissipation. The highest predicted power dissipation from high-performance microprocessor processors would reach around 360 W by 2020, according (Murshed,2017) to a research from the International Electronics Manufacturing Initiative (iNEMI) Technology Roadmap. In reality, the micro- and power-electronics sectors are tasked with removing a high heat flux of about 300 W/cm2 while keeping the temperature below 85°C. Furthermore, as devices get more integrated, power dissipation on the chip or device becomes increasingly non-uniform, as a peak chip heat flux might be several times that of the average. Of the environment

To deal with the enormous cooling demand and thermal management difficulties of developing electronic gadgets, traditional cooling methods are rapidly failing. As a result, in order to maintain their normal working temperature, high-performance chips or devices require new methods, mechanisms, and coolants with high heat transfer capabilities. Their usual performance and durability might decrease faster than expected if they are not adequately cooled. In addition, as the operating temperature rises, so does the rate of failure of electronic equipment. Their usual performance and durability might decrease faster than expected if they are not adequately cooled. Furthermore, when the working temperature rises, the failure rate of electronic equipment rises. Microchannel-based forced convection and phase-change cooling (liquid) are among the most promising approaches for attaining very high heat removal rates, according to reviews and assessments of research and progress of traditional and new cooling technologies. As a result, the major goals of thermal management have become effective heat flux removal and uniform power dissipation. Temperature management is (Murshed,2017) necessary to ensure that electrical equipment performs to its full capacity while remaining within design parameters. When a system's components are at risk of overheating, the system's operational capacity is jeopardised, and the system may fail catastrophically. While dealing

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with this design problem, electronics cooling emerges as a crucial approach for cooling components within electronic systems to prevent overheating and ensure performance. For example, to tolerate a heat flux of 100 W/cm2 at a temperature differential of 50 K, an effective HTC of 20,000 W/m2K is required, which is often not attainable by free and forced convections of these coolants. As a result, finding cooling fluids with better heat transfer capability is always a pressing necessity. As a result, there are a number (Sohel,2016) of new fluids on the market that might be utilised as sophisticated coolants. Nanofluid is a novel type of heat transfer fluids that consists of nanometre-sized particle suspensions in traditional heat transfer fluids such as water (W), ethylene glycol (EG), oils, and W/EG. When compared to their base conventional fluids, nanofluids were shown to have much higher thermal characteristics, notably thermal conductivity and convective as well as boiling heat transfer.

With highly desirable enhanced thermal properties, this new class of fluids can offer immense benefits and potentials in wide range of applications including cooling of electronics and other high-tech industries. There is a new class of fluids called "ionanofluids." Ionanofluids, which are nanoparticle (Bahiraei,2018) suspensions in solely ionic liquids, have also been discovered to have better thermal characteristics than their base ionic liquids. Ionano fluids have a lot of promise as sophisticated heat transfer fluids in cooling electronics because of their unique properties like being green fluids and being designable for specific jobs.

Many practical applications show a connection between natural convection and conduction heat transfer. The buoyant force generated by the temperature difference causes fluid flow in natural convection. As a result, no external force, such as fans or coolers, is required, which may otherwise be the source of noise and vibration. The heat transmission from the heat source to the fluid might be aided by the conductive solid (Sohel,2016) wall. The major disadvantage of natural convection is that it transfers heat at a slower pace than induced convection. Previous studies have shown that this problem may be solved by submerging the apparatus in a dielectric liquid.

If correct cooling technique is not utilised, component lifespan might be shortened, and the electronic device will not operate at its optimum. Failure to properly cool electronics and microprocessors, on the other hand, can lead to energy loss as well as system failure or destruction. As a result, high-performance microchips and electrical devices of various forms, sizes, and degrees of capability require (Riehl,2016) distinct thermal design techniques. Furthermore, coolants with high heat transfer capacities are necessary to increase the heat removal rate and, as a consequence, keep operating temperatures within acceptable limits. Heat

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pipes are passive, conductive, two-phase heat exchangers that offer significant thermal advantages for cooling high-performance, high-density electronic systems.

Passive cooling, which enables in their base heat sinks high latent heat fusion, high specific heat, controllable temperature stability and phase change materials (PCMs), is a relatively new technology used in transient (Riehl,2016) electronic cooling systems. A microscale cooling system cools the high heat generated by electronic devices. In electronic systems, 4 main mechanisms of thermal cooling are shown in the order of cooling demand. Coolants with high heat transfer capacity are essential to improve the heat capture rate and maintain normal operating temperature.

Altasim has experience in solving complex electronic cooling problems with the natural forces of convection in the environment. Natural convection for electronic cooling is based on the fact that the air is driven by heat that rises upwards and pulls down in cold air. Controlling the discharge of thermal energy requires the use of several approaches to obtain an effective solution. There is an urgent need to find coolants (Castro,2018) with superior heat transfer capacity. At One or more of these is to equip semiconductor chips with tiny devices in which a liquid is transported in micro-channels that pass through them to transport heat. To address the problem of high energy and the need to pump water-cooling fluid through tiny channels, researchers have developed distribution networks that narrow wide channels to precisely the places where heat is concentrated.

As new high-performance military aircraft become more and more dependent on the need to remove the thermal energy generated by increasing the performance of avionics packages. For example, tanks and crew carriers generate a significant amount of heat from weapons, communications and control systems that exceed the ability of air conditioners to remove them in hot desert environments. When the temperature of the low temperature reservoir is higher than ambient temperature, such devices will eject heat from the computer case and numerous other heat-generating devices by heating (Sahoo,2016) the surrounding air. The high thermal conductivity of the metal in combination with its large surface area leads to a rapid transfer of heat energy to the cooler air around it. Electronic cooling is surrounded by a low-temperature retention basin with a low surface temperature of chip heat recovery, allowing heat to be transferred to the chip without active cooling. Fans improve the transfer of heat energy from the heat sink to the air.

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The main *objectives* of this thesis is mentioned below

- To work on topic how importance the thermal optimization in electronic circuits. For this a detailed study on major works on the field of electronic cooling, types, factors affecting the lifespan of electronic component.
- Build a model circuit and analyse the thermal efficiency of the model.
- Analyse the model in natural convection condition and forced convection condition.
- Carry out migration methods in order to reduce the overall temperature of model geometry.

The below are *tasks* to be carried out in order to meet the objectives of this thesis

- Selection and creation of geometric model of circuit to be analysed for thermal efficiency with dimensions of electronic components similar to market available sizes. Here in this case is a simplified version of H-bridge circuit which is widely used in motor controller circuit is selected due to the growing importance of electric automobiles and reliability of controller boards is an inevitable factor as it is one of the main component of electric cars after batteries.
- After creation of model meshing operation the model would be carried out in Ansys Icepack. Setting of initial conditions of model for simulation, here in this case model simulation would be carried out in natural convection condition with flow type as laminar and forced convection condition with flow regime as turbulent condition was selected.
- Finally, interpretation of results would be carried out in 4 stages: thermal efficiency of model without any mitigation measures, with heat sink as remedy, with optimized heat sink and finally with optimized heat sink and fan.

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2. THEORETICAL FRAMEWORK

This chapter presents an overview of current research in the topic of electronic component cooling. A great number of work on the thermal management of electronic equipment has been published over the previous century, and many of these publications have been evaluated in this study, which has supplied valuable information in the area. Over the last 50 years, electronic gadget usage has surged, and they have become enmeshed in every area of our everyday lives. As the power and functionality of these devices evolved, new difficulties emerged, including thermal regulation of these electrical components and systems.

When the first electrical digital computers were constructed in the 1940s, excellent heat evacuation was vital to guaranteeing the steady running of later generations of computers. For optimal performance and efficiency of electronic equipment, the temperature of the components and the surroundings (Naphon, 2012) must be maintained or decreased. A typical example of contemporary CPU operating temperature fluctuates between (50-100°C) depending on the manufacturer and the intended purpose, according to Naphon et al. (Naphon et al., 2012). (Naphon et al., 2012).

2.1 Importance of Cooling in Electronic Systems

Among computer systems' electronic boards are the Central Processing Unit (CPU), motherboard, and RAM. These electrical components generated more heat as computers became more powerful and multi-tasking. If a maximum temperature is kept for components, circuit density complicates thermal management. If the maximum temperature of a component is to be kept below or at a given value, a higher circuit density complicates thermal management. When a silicon chip's temperature is beyond the permissible range, its reliability decreases by around 10%. (Bar-Cohen et al., 1983). Approximately half of all electronic equipment failures can be attributed to temperature-related factors, according to US Air Force research. Therefore, a cooling system that is efficient is necessary to avoid premature failure. Cooling and ventilation capabilities of the unit must also be improved (Yeh, 1995) in order to disperse excess heat and keep the components cold.

Electronics cooling applications require more flow and heat transfer than heat exchangers and are as challenging as cooling gas turbine blades (Moffat, 2002). The amount of heat generated by central processing units, in particular, can be concerning when there is not adequate cooling and heat dispersion. In addition, the excess heat generated can interfere with

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the proper functioning of the central processing unit and cause component and circuit malfunction. PC electrical components can't function properly without an adequate cooling system. A CPU that is overheated and not maintained at a safe temperature may result in loss of data and irreversible damage to a system.

2.2 Electronic Thermal Management

A number of cooling systems are used to remove heat from heat sources and their surroundings. There are four different types of cooling techniques:

1. Forced convection.
2. Phase change cooling.
3. Edge cooling.
4. Natural convection.

There has been a lot of research in these areas. When using one of the aforementioned cooling approaches, the study comprised a mathematical, computational, or experimental analysis that resulted in improved and more controlled heat transfer characteristics. Computational Fluid Dynamics models have not been used in the electronics cooling industry for a long time due to the expensive cost of performing CFD calculations. As a consequence of the introduction of high-performance workstations and personal computers, the cost of such computations has fallen dramatically (Kondo, 1998).

In today's world, forced air cooling systems are often employed to remove excess heat from computer CPUs. Because large and powerful fans are required to deliver the required air, the forced air cooling system is not suitable for many (Rodgers, 2003) high-dissipating applications. These fans may cause unwanted vibrations and noise, which might be harmful to the system. Furthermore, even if the CPU generates just a little amount of heat, the fans are always running at full speed.

More innovations in heat transfer technology will be required as the power requirements of multi-tasking electronic devices expand. As a result, more sophisticated and clever heat removal methods are necessary. Many high-efficiency cooling systems, such as liquid cooling or refrigeration, are being developed, however they have certain downsides, such as dependability, cost, and technical packaging issues. The next research will focus on electronic component thermal management, with a focus on sophisticated single phase fluid cooling systems.

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2.2.1 Natural Convection Cooling System Using Single Phase

Natural convection is the most efficient and cost-effective cooling method for electrical equipment (Ridouane et al., 2005). Furthermore, since it has no moving components and relies on the buoyant force caused by temperature differences acting on the fluid, natural convection heat transfer is the preferred approach for electronic cooling. In most situations, electronic equipment is kept cool via direct air cooling, indirect air cooling, or liquid cooling inside enclosures. For indirect single phase cooling systems within enclosures, there are two kinds of heat removal methods: forced convection and natural convection. It also makes a difference whether the working fluid is a gas or a liquid. Working fluids such as air, water, and dielectric liquids have all been employed in past studies. Heat transmission via natural convection from a heater that employs water and air inside a closed enclosure is investigated in this study, with the water being utilized since it does not come into direct contact with the electronic component. As a consequence, the utilization of water inside the enclosure for natural convection heat transfer is given considerable attention. Many practical and theoretical studies on natural convection heat transfer within the enclosure have been conducted in order to assess the capacity of natural convection schemes in comparison to the best thermal design for a cooling system of this kind. Air Cooling System:

Because air is readily available and does not need a closed system, it is the finest fluid for cooling electrical equipment. Forced air is often employed to eliminate the device's produced heat. Although forced convection is used within the computer case, the emphasis of this study is on natural air convection. The numerical analysis of a 2-D continuous natural convection in an enclosure (Aydin et al.,1999). The enclosure's left vertical wall is kept isothermal and cooled by the top wall. The remaining walls are insulated, and air is used as the working fluid. On fluid flow and heat transmission, the effects of Rayleigh number and aspect ratio were explored.

Small Rayleigh numbers (the ratio of the strengths of thermal transports due to buoyancy to thermal diffusion) have no effect on the average Nusselt number (the ratio of convective to conductive) when the aspect ratio (the ratio of enclosure length to height) is 1, and high Rayleigh numbers are largely reliant on high Rayleigh numbers. The average Nusselt number grows as the Rayleigh number climbs for square and shallow enclosures, and a significant portion of the enclosure is filled with cooler fluid, particularly for large Rayleigh numbers. In a square air-filled enclosure, (Radhwan,2000) investigated numerically steady state and 2-D natural convection heat transport. The cage's remaining walls were insulated, and one

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side of the enclosure was invisibly heated to a high temperature. The rest of this side, as well as the opposite side, were used as isothermal heat sinks. The Rayleigh number, which represents power supply changes, was raised from 102 to 106, and the effects of this change on temperature and flow fields for various heater locations were studied.

The results showed that the best spot is in the middle of the wall over a wide range of Rayleigh values. In addition, when the Rayleigh number increases, the eddy cell expands in size. The scientists discovered that their work had a greater heat transfer rate than an adiabatic vertical wall with a single separate heater. Another study looked at three-dimensional natural convection in a horizontal air-filled container with a discrete flush-mounted rectangular heat source at the bottom in a quantitatively stable condition (Sezai,2008). The source's aspect ratio (source length to enclosure height) was modified until it completely filled the bottom plate's width. Furthermore, the Rayleigh number fluctuated from 103 to 103 until an unstable flow was projected. Except for the chip, the bottom wall surface is adiabatic, while the top wall surface is kept at a constant temperature. Two types of boundary conditions were studied for the enclosure's sidewalls: (1) adiabatic and (2) constant temperature. Because the enclosure height is modest, they found that the vertical wall boundary conditions had no effect on the rate of heat transmission. Furthermore, the maximum Rayleigh number required for a convergent solution reduced as the aspect ratio of the source rose. Furthermore, the isolated heater's heat transmission was greatest near the borders and least in the centre. As the heater's length increased, the edge effects became less noticeable.

In a cavity with a heated plate positioned vertically and horizontally at multiple positions inside the air-filled enclosure, numerically unstable condition and 2-D natural convection were investigated. For varying aspect ratios of the heated plate, many Grashof values ranging from 103 to 105 were explored (Nithyadevi,2007) (ratio of height of plate to enclosure). The hollow's top and bottom walls are adiabatic, and the two vertical walls are always colder than the plate's temperature. When the Bossiness approximation was used, the flow was laminar and incompressible. As Gr increased, the heat transfer rate increased in both the vertical and horizontal locations of the plate, as seen by the data. Furthermore, when the aspect ratio of the heated plate was decreased, the heat transmission rate was reduced. Experimental and computational studies of air natural convection in a slanted square cavity were conducted. Isothermal hot and cold walls were created by controlling the temperature differential between two walls and separating the other walls. The Rayleigh number, which shows the power input, was examined at various levels.

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The flow was affected more in regions near active hot and cold barriers, according to their findings. In convection exchanges, the cavity's inclination was crucial. Furthermore, the exchange rate varied significantly depending on where the hot wall stood in relation to the cold wall (more or less raised). The numerical simulations were found to match the data well, as seen by the tiny deviation figures, with the largest difference being about 12% for high Rayleigh numbers. Deng investigated laminar natural convection in a two-dimensional air-filled square cavity with a large number of discrete source–sink pairs flush installed on the vertical sides. To account for thermal buoyancy effects, the Boussinesq approximation is applied. They looked explored how the size and position of each source and sink affected the flow characteristics of multiple cells in an enclosure, and hence the overall heat transfer (Deng,2008). The heat sources were kept at a greater temperature than the sinks throughout the simulation. The enclosures' remaining components were also thermally insulated.

When the layout was altered from separated to staggered, the buoyancies of the sinks and sources dissolved, the number of eddies in the enclosure grew, and therefore the heat transmission increased. As the number of eddies rose, the vigorous mixing and heat transfer between them resulted in a more uniform temperature of the fluid in the enclosure. As the sources and sinks were broken into smaller pieces, the number of eddies in the enclosure grew, improving heat transmission. Sudhakar et al. investigated 3-D free incompressible and laminar air convection heat transmission in vertical channels with numerous protrusion heat sources subjected to uniform heat generation using both experimental and computational approaches. When one vertical wall was believed to be adiabatic, the goal of their investigation was to reduce the component temperature to a value within its approved (Sudhakar,2010) limits. In both the theoretical and experimental studies, the duct and heat sources were the same size. The pressure and velocity equations were linked using the SIMPLE technique, and the controlling flow and energy equations were solved using the commercial tool FLUENT. Different hot module sizes and channel spacing values were used in the testing. With a maximum variation of 10%, the researchers found that the computational and experimental temperature data were in good agreement. It was revealed that the location of the heat sources has a substantial influence on fluid flow and heat transfer characteristics for the same amount of power.

Bari et al. (Bari et al., 2014) investigated natural convection in closed air-filled cavities using unstable numerical and experimental data. The electronic gadget's hot wall was made up of three parallel discrete bands that provided a continuous heat flow and were separated by two equal-length adiabatic bands. The hot and cold walls were kept vertical, and the cold active wall

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of the enclosure was kept isothermal. Several different designs were investigated for a wide variety of Rayleigh numbers, including high values and varying degrees of inclination. The experimental component of the study was carried out using a set-up based on electrical data and wall temperature measurements. Natural convection is preferred for positive angle values when the hot wall is lower than the cold wall, according to the authors. When the hot wall was entirely active, as opposed to having distinct active areas, natural convection was improved. The temperature values derived from the numerical simulation were pretty similar to the real data, despite the fact that the numerical research was done in two dimensions.

To remove heat from the chip surface, modern desktop computers employ direct or indirect air/liquid cooling. Natural convection cooling inside rectangular enclosures for cooling electronic chips has shown that poor convective heat transfer coefficients impeded air cooling. The conventional method of employing free convection to remove the large amount of heat created by contemporary electronic chips (Bari et al., 2014) is no longer sufficient to remove the vast amount of heat generated by those modern electronic chips.

Using liquids instead of keeping electrical equipment at a safe operating temperature. This is due to the fact that liquids have far greater thermal conductivities than gases, resulting in much larger heat transfer coefficients. As a result, liquid cooling almost always beats gas cooling (Etemoglu, 2007). Indirect liquid cooling, on the other hand, necessitates the installation of a distribution system that comprises a pump and plumbing connections. Because of undesirable noise and vibration induced by fan or pump components, future desktop PCs may be ruled out. To address these issues, liquid natural convection inside an enclosure offers a viable approach for draining heat from computer chips while maintaining the proper temperature.

2.2.2 Liquid Cooling System

Conjugate natural convection with liquid has attracted a lot of interest since it is a simple and cost-effective alternative to forced convection for a variety of applications, including the construction of efficient heating systems and the cooling of electronic equipment. Natural convection cooling has been and continues to be one of the most promising methods for controlling the temperature of electrical equipment. Natural convection currents form when the heated and cooled fluids are switched on a regular basis. Because the density of the fluid changes, natural convection heat transfer occurs when a light fluid is replaced with a heavy fluid (Cengel, 2003). Because temperature gradient changes drive fluid motions inside

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enclosures (whether these enclosures were vertical or horizontal) in natural convection, the isothermal boundary condition is used in most research. A variety of experimental and computational investigations on liquid natural convection heat transfer have been published.

Sathe and Joshi quantified the heat transfer and natural convection caused by a heat source placed on a substrate and submerged in a 2-D fluid-filled enclosure. The substrate and heat source were vertically positioned in the enclosure's middle. The effects of varying Rayleigh numbers, Prandtl numbers, and the thermal conductivity of the substrate/fluid ratio were explored by (Joshi, 1991). The Rayleigh number is proportional to the quantity of heat flow, according to their results, and is represented as R a=gq" h4/vk, where h stands for height, kinematic viscosity, thermal diffusivity, and volumetric thermal expansion coefficient. Raising the Rayleigh number improved heat source cooling, they observed. When the solid temperature was between 20°C and 30°C, the change in Prandtl number Pr had no effect (10:103). The amount of heat that passed through the substrate rose as the value of the substrate/fluid ratio Rs increased. When the Rs = 103, the substrate lost around 80% of its output power. They claimed that 2-D models might provide a good approximation of component surface temperature in different electronic packages.

In an experimental environment, Keyhani et al. investigated the influence of the rectangular enclosure's aspect ratio (ratio of enclosure length to heater height) on natural convection heat transfer. On one vertical wall, five projecting heaters were attached, and the opposite wall was adjustable, allowing the enclosure width to be modified (Keyhani,1991). The enclosure's top surface acted as an isothermal heat sink, while the rest of it was insulated. The cage's convective medium was ethylene glycol. Six different cavity width values were used in the tests. They observed that the width of the enclosure has an influence on heat transfer, but that this effect is restricted when the enclosure width to heater height ratio is about 4.0. In a square enclosure filled with water, (Sathe,1992) investigated numerically 2-D and steady conjugate natural convection heat transfer from a protrusion placed on the substrate (FC-75). For a range of Rayleigh numbers, enclosure boundary conditions, and protrusion thermal conductivities, the solution was found. They observed that the relevance of substrate conduction decreased as the Rayleigh number increased. With the exception of values less than 100, where there was minimal change in the maximum temperature level, the maximum temperature in the chip fell as the substrate/fluid thermal conductivity ratio increased.

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According to studies, the temperature differential between two plates must surpass a particular threshold before the first symptoms of heat transfer and fluid motion may be seen when a fluid is contained between two plates. It used a heated chamber for continuous natural convection to run 2-D and 3-D computational projections as well as real trials. They ran numerical simulations with two distinct working fluids, water and FC-77, given the same geometry of the tests (Jouhara,2010). At room temperature, the distinct heat sources on the vertical substrate and the opposite wall of the rectangular hollow were both isothermal. The researchers noticed that raising the thermal conductivity ratio between the substrate and the fluid lowered the amount of energy delivered via the heaters to the fluid while increasing the quantity entering the substrate. As a result, along the solid/fluid boundary, the local dimensionless temperature drops.

Aminossadati investigated 2-D steady state natural convection in an isosceles triangular enclosure filled with Ethylene Glycol–Copper nano fluids and heated horizontally at one of its walls (EG-Cu). While the two other walls were considered to be isothermal, they numerically examined the impacts of solid volume fraction, variable heat (Kargar,2011) source position, and Rayleigh number on thermal performance. They found that increasing the Rayleigh number and solid volume percent increased the enclosure's thermal performance. The heat transfer rate was affected by the position of the heat source, which had varied Rayleigh numbers.

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2.3 Common Electronics Design Obstacles

2.3.1 CAD & Geometry

It's not uncommon in electronics cooling design to spend a large amount of time developing CAD models for simulation. Most simulation tools will reject detailed production-ready models in their current condition, forcing simulation engineers to go through the arduous process of de-featuring geometry, simplifying sections, closing minor gaps in the model, and repairing erroneous geometry. And, in the case of geometry studies involving several versions of a single base model, a lot of this work has to be done again. Traditional CAD tools in simulation software are inefficient or ill-equipped for such tasks in many circumstances, as they are designed for creating new models rather than simplifying old ones.

2.3.2 Design Efficiency

The inability to swiftly explore and replicate design performance is a prevalent issue in early-stage design. Engineers seek the opportunity to receive feedback on their designs in an iterative way, employing fast and precise thermal and flow analyses. Lack of access to engineering simulation during the early stages of product development can have a substantial impact on product performance.

2.3.3 Cost

Traditional simulation stacks typically require large upfront investments before demonstrating the added benefit of all acquired gear. With such comes a great deal of responsibility in terms of system maintenance, as well as a significant know-how barrier that necessitates teams doing specialized training. Integrating cloud-native simulation into thermal management design has the potential to significantly reduce product development costs. The less expensive design research is, the faster it converges on its final version. Designers can use simulation tools to discover a problem before spending money on actual prototypes.

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2.4 Techniques for Controlling Temperature

Heat can be transferred through electrical systems in a number of ways, as shown in the previous section, and if the temperature climbs too high, it can have a substantial influence on the system's function. As a consequence, it's desirable to avoid overheating an electrical system. The maximum operating temperature of its key components In the design of electronic systems, there are numerous solutions for reducing the impacts of temperature. Some of the packaging choices include passive thermal management, active thermal management, and advanced electronic thermal management.

2.4.1 Heat Sinks

The most popular passive thermal management structures are heat sinks. Heat sinks are thermally conductive structures that expand a heat source's effective surface area. To improve the amount of surface area from which thermal energy can be dissipated, these devices often employ a large number of equally distributed, thin (Gururatana et al., 2012) fins. H. Mousavi et al. previously demonstrated that irregular structures like interrupted, staggered, or caped fins can improve heat sink efficiency by reducing the maximum temperature of a system with a 2000W source by up to 6.7°C. S. Gururatana et al. came to the same conclusion using interrupted and staggered elliptic fins. Tsai et al. tested a bead-based heat sink against a 65W CPU and were able to lower the highest surface temperature by 39oC, or 43%. A water-cooled dimple and pin-fin heat sink, can lower the temperature of a high heat flow.

Density source has increased by as much as 15 percent. The efficacy of a dimpled, cylindrical shell heat sink was tested by Zheng et al., who discovered that it could lower the mean wall temperature by up to 60 degrees Celsius. Chaudhary et al. demonstrate that adding wings to a common pin fin heat sink improves heat transfer rates across the board. Regardless of design approach, the fundamental underpinning heat sink design remains the same: increasing the surface area of a structure decreases its equivalent thermal resistance (Zheng, 2017), hence boosting heat dissipation. As a consequence, a component is unable to store a large amount of thermal energy. The classic finned or pin heat sinks are still the most prevalent forms of heat sinks, even in high-performance applications. This may be attributed to their basic design when contrasted to more sophisticated structures such as dimpled or cylindrical-shell heat sinks.

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2.4.2 Thermal Vias

Thermal Vias are used in substrates and electronic packages to facilitate heat transfer through thermally resistant, low heat capacity materials. The three most prevalent varieties of thermal vias are hollow shell vias, filled shell vias, and solid core vias. Regardless of the route, they operate as thermally conductive conduits in an otherwise thermally refractory material. Thermal vias are organized into dense, evenly dispersed square-shaped arrays called as Thermal Via Arrays by packaging experts to adjust the equivalent thermal resistance of electronic packages (TVA). As a consequence, heat generated on one side of the substrate is distributed to the other side. This prevents a large amount of heat energy from being stored underneath active components.

This reduces the effects of temperature coefficients on the system, as well as the likelihood of temperatures exceeding critical levels such as junction, phase-change, and glass-transition temperatures. Because of their simplicity, via-based heat management mechanisms are especially common in electronic packaging systems. Shen et al. performed detailed parametric calculations in one and two dimensions to better understand the design space of thermal vias. Shen et al. recommend that via-to-via spacing be kept to a minimum, that vias be filled with conductive material, that the via diameter be chosen based on the thermal conductivity of the via fill material, and that additional planar metallization layers have no effect on the PCB's through-plane thermal resistance (Catalano,2020). Shen et al. showed that there is an optimal through diameter for through-plane thermal conduction that minimizes the equivalent thermal resistance. By applying these results to a thorough design technique, Shen et al. were able to achieve thermal resistance values as low as 36 Wm-1K-1.

Another kind of thermal via is one that passes through silicon. Even though these thermal vias are microscopic solid-core structures placed directly into an integrated circuit, the physics and design philosophy are almost identical. TSV research has focused on decreasing total metallization (to eliminate routing challenges) and lowering thermal stress, while most thermal vias research has focused on lowering through-plane thermal resistance for improved power dissipation. According to Kuo et al., a CTE mismatch between a TSV and the surrounding material may induce compressive (Xiao,2015) stress when heated. This stress is greatest when the heat source is near to the thermal outlet and reduces as the distance between the heat source and the thermal outlet increases. According to Xiao et al., circular flaws have a 16 greater influence on thermal stress than square or triangular defects, and the highest stress

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between the TSV and the surrounding structure occurs at the interface between the two materials.

Zhang et al. claimed that increasing the diameter to length ratio of a TSV may lower equivalent stress in a silicon body, mirroring Xiao et alobservation .'s that thermal stress is greatest at the interface between the via and the surrounding host material. According to Takao and Taya's previous work, embedded inclusions (cylindrical or otherwise) may cause thermal stress-induced cracking and delamination in their host matrix even when the temperature rises uniformly. In addition to measuring the stress generated by TSVs, Goplin et al. noted that effective through-silicon thermal via placement may be employed to decrease both the (Zhang,2014) maximum and average of the thermal gradient. The researchers discovered that adopting smaller, homogeneous via arrays and reducing the number of vias in strategically located through zones may effectively minimize the maximum and average temperature disparity. Additional metallization has a detrimental impact on the maximum and average temperature gradients, which level off to asymptotic behavior when metallization accounts for roughly 48% of the volume.

2.4.3 Mass Via Arrays

MVAs are a heat-dissipation enhancement to TVA's electronic packaging technology. Thermal vias are used in both of these technologies to increase heat transfer efficiency in three-dimensional structures. In these cases, thermal vias are utilized to reduce equivalent thermal resistance by building thermally conductive channels in a device.

MVAs and TVAs are both made up of a dense rectangular array of thermal vias. MVAs and TVAs have similar fundamental architecture, but their design philosophy is vastly different. MVAs and TVAs both strive to lower a substrate's equivalent thermal resistance, but MVAs' design philosophy also tries to modify thermal energy flow in nonlinear ways by adding intermediate heat spreader layers and strategically arranging individual vias. Heat (Sim,2012) spreaders improve the structure's thermal resistance while lowering the equivalent thermal mass, allowing for more efficient heat dissipation and a lower maximum temperature. MVAs and TVAs vary not just in design concept, but also in design execution. Instead of being a periodic 2-D structure, an MVA is thought of as a stack of n-many independent TVAs. Due to their three-dimensional character, MVAs may have both internal and exterior boundary conditions. If MVAs are implemented in a technology that allows for the construction of internal micro-channels, such as Low Temperature Co-Fired Ceramic (LTCC) substrates, both

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internal and convective boundary conditions may be applied. While TVAs and micro-channels have been shown to be effective for convective heat transfer, adding interlayer heat spreaders and selective through placement to MVA systems may increase heat dissipation effectiveness. The interlayer heat spreaders in MVA structures act as heat sink fins, increasing the surface area of the boundary and enabling for more effective heat dissipation than TVAs.

2.4.4 Liquid Cooling

Fluidic cooling is a kind of forced convection cooling in which thermal energy is transferred away from a heat source to a heat sink by a fluid, which may be either a liquid or a gas. Fluidic cooling systems are usually effective in dissipating heat, but they are often large, bulky, and unsuitable for small-scale applications. Microfluidic cooling is a kind of fluidic-based cooling that is better suited. Traditional fluidic cooling systems employ micro-fluidic behaviour to disperse heat on a much larger scale than micro-fluidic cooling. The efficiency of such systems has been the subject of several studies.

Microchannel with a high flow rate were able to lower the maximum temperature of a structure by more than 4oC when compared to a system with no fluid flow, according to Hu et al. Zhang et al. discovered that utilizing microfluidic channels with incorporated passive features, such as thermal vias, they were able to maintain a maximum temperature of 750C while cooling a 100W source. Microfluidic-based cooling systems using thermal vias are an extraordinarily (Zhang,2017) efficient small-scale cooling system, according to Lee et al study, and the number of thermal vias should be increased for optimum heat dissipation. Liu et al. demonstrated the use of thermal vias in combination with micro-channels for improving heat transfer rates; these structures were able to lower the maximum temperature induced by a 30W source by roughly 40%. I-shaped micro-channels implanted into a multi-layer LTCC substrate may lower the substrate's maximum temperature by 73.4 percent, according to Miao et al. According to Pietrikova and Girasek's study, integrating microchannel onto multilayer LTCC substrates enhances the capacity to drain heat from active electrical components. After introducing a variety of fluidic structures into LTCC, it was discovered that a broad wedge-shaped channel was more effective at dispersing heat than other meandering patterns. The broad, wedge-shaped channel lowered the equivalent thermal resistance of the substrate by 2.8 percent and was able to lower the temperature of the electronic test vehicles by 86 percent when compared to previous meander designs. Bar-Cohen et al. showed that certain high flow rate microchannels caused an increase in system temperature and suggested that jet impingement cooling devices be utilized instead. The investigation According to Deng et al., impingement

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jet cooling might reduce the maximum temperature. The working temperature of a laser diode array is increased by 5.38oC. According to R. Wuet al's study, a 500W device can be cooled by up to 35oC utilizing a jet impingement body cooling system as compared to a device that isn't cooled.

A great amount of study has gone into the optimization of these structures. These studies reveal that variables such as fluid flow velocity, fluid thermal conductivity, channel geometry, channel wall thermal conduction, surface roughness, and others may have a considerable influence on the efficiency of these devices (Zhang, 2017). These same articles have shown that, although these cooling approaches may reduce the temperature of a high heat flux density system, they typically need exotic materials that are not suitable for large-scale application and perhaps challenging manufacturing procedures. Integration of very tiny channels into multi-layer substrate systems, such as LTCC, may be a tough challenge in and of itself

.2.4.5 Thermoelectric Cooling

The thermo-electric phenomena was discovered in the 1790s by Alessandro Volta, an Italian scientist. This phenomenon proved that a heat gradient at the interface of two distinct but complimentary materials may result in an electrical current flowing between the junctions of the materials. A temperature gradient may be caused by an electrical current at the junction of two materials, as has been proved. This phenomenon has caught the interest of several academic and industrial fields because of its capacity to transform waste heat into electrical energy or produce a temperature gradient that may be exploited for solid-state refrigeration or heating (Zhao,2014).

For this application, thermo-electric coolers are an excellent match (TEC). The Peltier effect is used by thermo-electric coolers (TEC) to promote heat flow when an electrical current is created in the junction between two different materials. As a consequence of the heat flow, the gadget has a cold and a hot side. The bulk of these devices will be electrically parallel doped structures with thermally conductive ceramic layers above and below. While these TECs have been investigated for usage in a variety of current applications due to their ability to create cooling heat fluxes of up to 10 W/cm2, they are severely restricted, since conservative roadmap (Zhao, 2014) projections indicate average chip heat flux will soon surpass 150 W/cm2. As a consequence, academia and industry are experimenting with small silicon TECs, such as contact bumps, to see whether they can provide a significant heat flux for sub-mm hot spots.

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According to a parametric study by Luo et al., reducing the aspect ratio of each thermoelectric leg may improve thermoelectric cooler performance while having a little influence on the device's electrical efficiency. Specifically, raising the aspect ratio from 0.6 to 1.8 increased the coefficient of performance by 118.7% while only increasing the electrical inefficiency by around 1%. According to Chowdhury et al's study at Intel, in-plane silicon TECs, in addition to passive cooling structures, were able to lower sub-mm hot spots by as much as 14.9oC. The heat produced by a 10.8W thermal source may be reduced from 111.4°C to 54°C, according to Mahajan et al(Luo,2018) .'s experimental assessment utilizing commercial thermoelectric models. Saber et al. created a three-dimensional thermoelectric-based cooling system that employed a single-stage TEC to decrease the heat produced by a 15W source by up to 25.5 degrees Celsius. Combining gravity-assisted heat pipes with TECs, according to Sun et al., may enhance cooling capacity by 64.8 percent while increasing power efficiency by 39.3 percent.

Ahammed et al. demonstrated that by employing a nanofluid-cooled, multiport, mini-channel TEC, they were able to boost a power transistor's cooling efficiency from 19.22% to 23.92%. Electrically, TECs have a substantial disadvantage; the TECs used by Chowdhury used 3A, but typical plate coolers may need greater than 20A. A second difficulty with TECs is the need to cool the hot side of the device. Because most cooling systems (Ahammed,2016) do not need an additional cooling strategy to work, this is a distinct difficulty. Yang et al. discovered that a variety of secondary cooling solutions were used with varying degrees of efficacy, including air, water, microchannel, PCM, and heat pipe cooling. These challenges just add to the hardships that people from are already facing.

Regardless, the only way to cool a particular section of a system to a temperature lower than the ambient temperature is to use thermo-electric coolers. The ability to modify the size of TECs is also particularly helpful since it can be used to manage 'hot spots' on circuit boards that might otherwise severely reduce an electronic component's life.

2.4.6 Electronic Packaging Improvements

It is well understood that introducing anisotropic features to an otherwise homogeneous medium may interrupt energy flow. This basic concept is one of the driving factors in the development of metamaterials in the fields of electromagnetics, thermodynamics, and structural dynamics. We're curious in how these anisotropic properties impact heat flow and, as a

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consequence, temperature distribution within a given region, in the context of thermodynamic systems, which is the focus of our study.

Anisotropy is used in nanostructured materials research to affect the bulk and surface behaviors of systems. By integrating nanoscopic fibres, such as graphene, into substrates or interface materials, packaging engineers have sought to redirect heat flow and reduce thermal impacts at the nanosomic scale. In a general fashion, Khodadadi exploited these structures to increase bulk material thermal conductivity. Phase Change Materials were produced by Khodadadi et al. using carbon-based nanostructures, metal nanoparticles, and nanowires (PCMs). When compared to the integration of metal nanoparticles (Khodadadi,2013), the findings of that study revealed that introducing carbon nanostructures into PCMs resulted in significant gains in bulk thermal conductivity. Similarly, Warzoha et colleagues investigated the use of carbon nanotubes and graphite nanofibers as a sort of Thermal Interface Material (TIM) between IC circuitry and packaging to minimize thermal contact resistance. A variety of films were produced to enhance the surface roughness between the heat source and the thermal outflow as part of this research. These TIM mats have been shown to beat commercial Arctic Silver alumina-based TIM by up to 81 percent in most tests. According to McNamara's review, while nanoparticle-based TIMs can reduce boundary resistance between different objects, the research is still in its early stages; manufacturing and assembly of nanostructured TIMs raises questions about their feasibility for large-scale (Warzoha,2013) application, and little research has been done to investigate their performance under harsh conditions. While nanostructured materials provide a wide range of thermal management choices, meso-scale systems provide many of the same benefits - especially in terms of assisting tiny electronic systems - while also improving manufacturability. Thermal metamaterials are nanostructured materials with a meso-scale variation. Thermal metamaterials may be created by intentionally mixing several materials with different thermal conductivities and specific heats to change the thermal energy passing through media. Thermal metamaterials may be employed in a system to shield, concentrate, and reverse heat flow. These techniques are used to reduce heat flow, concentrate heat flux density, and modify the direction of heat flux within specified parts of a system.

Hatta & Taya's thermal inclusion methods are used in the present state-of-the-art. Using these optimization approaches, Dede showed the ability to protect, concentrate, and reverse heat transport in arbitrary volumes. Heat flux shielding, concentration, and reversal were proven in this work's concepts, which were able to be incorporated into an actual system. Yang et al. illustrated how a thermal ground cloak may be made using a highly anisotropic construction

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with alternating layers of material. As a consequence of the study, an ultra-thin thermal cloak was developed, which could shield a particular area from an external heat flow and could be shown in both 2-D and 3-D situations. When compared to a pure copper heat spreader (Yang,2019), alternating layers of high and low thermal conductivity material in the plane of heat transmission may significantly enhance the spreading effectiveness of a heat spreader. Furthermore, the research discovered that increasing the thermal conductivity ratio of the alternating layers as well as thickening the heat spreader had a positive influence on the heat spreading efficiency of the structure.

Annealed Pyrolic Graphite (APG) has lately received a lot of interest due to its low ratio of in-plane to out-of-plane thermal conductivity. In the XY plane, thermal conductivities have been observed to vary from 400 Wm-1K-1 to 3500 Wm-1K-1. This phenomenon (Yang,2019) permits heat to be spread uniformly even when a localized source is present. As a consequence of this phenomenon, APG is widely employed in out-of-substrate heat spreader devices, in which the APG is enclosed in an aluminium shell and connected to the substrate by thermal vias. This structure, known as K-core by Boyd aerospace, enables for very efficient heat management in ultra-compact power electronics applications.

2.4.7 Heat Sink as Thermal Dissipation Method for Electronic Circuits

Heat sinks are devices used in high-power systems to disperse a significant quantity of heat produced by the system's components. A heat sink is used to enhance the surface area of the component, allowing it to disperse heat more quickly and keep it within the manufacturer's specified operating temperature ranges. The heat sink conducts heat to its surroundings (ambient) either passively or violently, depending on the working conditions and materials. Conduction absorbs heat, which is subsequently released into the environment by convection or radiation. Heat sinks are formed of thermally conductive materials like copper and aluminum and disperse heat by liquid cooling, natural convection, forced convection, or radiation (Aravindh, 2008). Convection, often known as radiation, is a process in which heat is transferred from one place

A heat sink is an electronic assembly or other heat-generating component that transfers thermal energy to a cooling liquid. Heat is transferred from a higher temperature zone to a lower temperature zone through conduction, convection, radiation, or a combination of these heat transfer techniques (fluid medium gas or liquid). The heat sink's performance is influenced by the velocity of the coolant fluid, the material's thermal conductivity, the thermal interface

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material, and the method of attachment. It's necessary to submit certain applications. They used simulation to evaluate and improve their performance. Recent breakthroughs in semiconductor materials, fluorescence methods, driver electronics, and thermal management, among other technologies, such as concentrated solar cells, have prompted the use of high-power light emitting diodes (HP-LED) for public lighting.

2.4.8 Necessity of Optimization of Heat Sink

The material selected and its composition are important considerations in heat sink design. Heat sinks built of composite materials with non-metallic elements may be a viable solution where performance, affordability, and manufacturability are important. The heat sink's performance is increasingly demanded as the amount of dissipated power per unit area of electronic components grows. A heat sink is a device used in computers to remove a significant quantity of heat created during operation by components such as integrated circuits (ICs), chipsets, and graphics cards (Culham, 2001). The heat sink is utilized to expand the surface area of the heat sink, allowing it to transfer heat more quickly and keep the ICs at a safe working temperature, which is done by tight design parameters.

Thermal management is critical for high power density in systems with all components, such as capacitors, inductors, and semiconductor devices, as previously stated. The manufacturer specifies the maximum working temperatures for these components used in power converters (Staliulionis, 2014). The surge in power density is the key cause that has pushed thermal management to the forefront and necessitated skilled control. According to (Ong,2017), thermoelectric (TE) devices may enable for clean energy conversion while also being ecologically beneficial. A thermoelectric generator is recommended for generating tiny amounts of electrical energy by using natural temperature variations between the air and the ground. Because even the greatest thermoelectric generators have a poor conversion efficiency, the performance of the heat sinks that create the heat flow is crucial. The performance of a thermoelectric (TE) module linked to a typical fin heat sink with a comparable sized heat source was investigated in an experiment.

Another study focused on high-power LED chips. To undertake a theoretical investigation and actual observations of temperature stability of high-power LED chips, the researchers employed an aluminum passive heat sink developed for use in a small light bulb configuration. The researchers showed that their method (Balvis,2015) maintained the temperature of the LED chip below 70 degrees Celsius, allowing it to function for extended

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periods of time. The simulations were conducted using a low-cost gadget that might be utilized for public illumination. Theoretical calculations and experimental data in many settings demonstrate a good match.

Concentrated photovoltaic solar cells were used to research solar energy. Cui (2009) describes the development of an energy-conserving thermal model for concentrator solar cells. The cell temperature would reach about 1200 °C if the solar cell was under 400X concentration and no cooling assistance was used. Metal plates were used as heat sinks to cool the system, resulting in a significant reduction in cell temperature. The cell temperature declined as the heat sink area increased at a constant concentration ratio. The heat sink area must rise linearly as a function of the concentration ratio to keep the cell at a constant temperature. To test the idea, GaInP/GaAs/Ge triple-junction solar cells were constructed. When utilizing a heat sink at 400X concentration, a cell temperature of 37 °C was recorded. Passive cooling costs and yearly energy outputs for high concentration photovoltaic (HCPV) systems were investigated in a separate study. In a system (Micheli, 2016) that has been placed in a variety of places across the world, the heat sink's efficiency was compared to the expected energy yield of a triple-junction GaInP/GaAs/Ge HCPV cell. Copper and aluminum heat sinks have been studied for their advantages and disadvantages. To get insight into a basic flat plate heat sink, the finite element analysis software tool COMSOL was used.

The form of the heat sink is also significant, according to (Soodphakdee,2001). The impact of heat sink geometry appears to have been investigated. The performance of numerous regularly used fin types in terms of heat transmission was investigated. At moderate laminar air velocities and pressure gradients, realistic, manufactural shapes were examined for decreasing thermal resistance. Plate fins or pin fins, which might be circular, elliptical, or square, are used. The plate's fins might be staggered or continuous (parallel plates). Pin fins may be positioned in a staggered or a straight configuration. A reference was provided in the form of Modeling and Experimental Evaluation of Single Component Passive Heat Sinks for Miniature High-Flux Photovoltaic Concentrators. The investigation was carried out both numerically and analytically. In the gathered analytical results, Bessel functions were identified, which need more than intuitive abilities and a lot of effort to attain.

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2.5 Other Related Works

In a vertical rectangular channel filled with water, Bhowmik et al. used four in-line flush mounted square chips to study convective heat transmission. The Reynolds number, they discovered, has a significant impact on the heat transfer coefficient. Maddox et al. conducted experiments on high-power electronic equipment using a variety of water jets. They built a connection to investigate the effects of Reynolds number, Prandtl number, and jet height, and discovered that increasing velocity increases critical heat flux (CHF), and increasing velocity further reduces the equipment's boundary layer thickness (Kumar, 2014). Tso et al. used four in-line flush mounted square chips in a water medium to perform an experimental single-phase forced convection study. They discovered that the temperature of each chip was a strong function of Peclet number, and they identified three unique heat transmission regions: laminar, transition from laminar to turbulent, and turbulent.

Wadsworth and Mudawar used dielectric Fluorinert FC-72 to investigate forced convection cooling of 3 3 square electronic chips (12.7 mm x 12.7 mm) and discovered that cooling the chips through rectangular jets maintained nearly isothermal conditions at the chip surface and that the chip cooling rate was independent of channel height. In a single-phase liquid cooling medium, Agbim looked into the experimental and numerical assessments of high-power density electronic components. Tuckerman studied the heat conduction from integrated circuit chips (q = 20 W/cm2) in a water cooling medium in an experimental study. Gupta and Jaluria investigated forced convection cooling for arrays of protruding discrete heat sources (4 3) using (Hotta,2020) determined heat input values in an experimental study. They discovered that the heat source arrays' average heat transfer coefficient did not change with different heat inputs. Two-phase cooling resulted in a 10% reduction in IC chip temperature when compared to single phase cooling. Gdhaidh used both computational and empirical methods to investigate single phase cooling of electronic devices in natural and forced convection media. The cavity size had a significant impact on the heat transfer mechanism and the maximum temperature of the heat sources, according to the researchers. Hotta et al. carried out experimentally driven ANN-GA based optimization studies for cooling discrete heat sources in mixed convection. The ANN-GA optimization strategy correctly anticipates the behavior of the heat sources, they discovered.

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Heindel et al. used numerical simulations to investigate 2D and 3D conjugate natural convection for a discretely heated cavity in water and FC-77 medium. They discovered that the distance between the heat sources on the board had a significant impact on the temperature of the board. A comprehensive examination of heat source cooling in a liquid media is presented by Patil and Hotta (Heindel, 2008). Water may dissipate up to 1700 W/cm2 of heat flux at a speed of 47 m/s, according to Oh et al. In a water medium, Casano and Piva evaluated the SMPS board's cooling performance. A large-capacity SMPS board was explored. To estimate the unknown heat flow in the conjugate heat transfer mode, Harsha et al. used a combination of ANN and GA optimization techniques. They found a good match between the predicted and actual heat flow, indicating that ANN might be useful for the time-consuming CFD model. Wei et al. studied liquid jet impingement cooling for high spatial resolution thermal performance of programmable chips in both practical and theoretical settings (8 8). (eight) Both outcomes are in agreement, they discovered. They've also suggested that single and multiple jet coolers have a link.

Natural convection is a fascinating and essential mechanism that occurs when buoyant force varies over space, causing fluid motion. These changes might be caused by changes in concentration or temperature, for example. The sort of natural convection described in our research is that caused by temperature variations resulting in differences in fluid density. Natural convection, which is closely connected to the temperature field, is caused by it. Fluid motion, due to heat convection, changes the temperature field (Patil,2020). Natural ventilation may therefore be used for convective cooling (as in electronics cooling) or fluid motion (as in pumping fluid through closed-loop channel systems). Structural optimisation is a traditional engineering field that entails modifying a structure's design to improve its performance in terms of a desired behaviour.

In the design of heat sinks for electronics cooling, size and configuration optimisation are common structural optimisation approaches. Morrison, for example, improves plate fin heat sinks in natural convection using a downhill simplex approach and empirical correlations. Morrison's main design considerations are fin thickness, fin spacing, and back plate thickness. Bahadur and Bar-Cohen optimize staggered pin fin heat sinks for natural convection cooled (Alexandersen,2014) microprocessor applications using mathematical models. The design components evaluated here include pin height, diameter, and spacing. Simple optimization techniques, such as changing the aspect ratios of reactor cylinders or the diameter, length, and

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configuration of closed-loop channel systems, have also been used to improve the design of buoyancy-driven polymerase chain reaction reactors.

Traditional optimization procedures may considerably improve present designs, but they are always limited in terms of design flexibility since a starting design must be defined in advance. This is where the topology optimisation technique shines, since it allows for the identification of previously unknown and surprising designs. Bendse and Kikuchi popularized topology optimisation as a material distribution approach for determining the best structural arrangement for a given issue with design constraints (Kikuchi, 2016). Simultaneously with the homogenisation technique, the density or SIMP (Solid Isotropic Microstructure with Penalisation) methodology, which is now the most used numerical method for topology optimization, was created. Bendse proposed the SIMP approach, which has been widely used by Zhou, Rozvany, and coworkers. Although the topology optimisation approach originated in structural mechanics, it has since been used to a wide range of fields such as acoustics, photonics, fluidics, and many more.

Temperature gradients and rises may damage electrical performance since they affect clock skew directly. Because joule heating connects electrical and thermal performance, a combined examination of the two is required to accurately estimate temperature distribution. The clock tree must also be simulated in the presence of the temperature distribution in order to estimate clock skew. Because many aspects such as physical geometries, interface materials, fan (Park,2017) speed, and others have a direct impact on the temperature profile, they must be adjusted properly to provide the best electrical performance. As a result, this becomes a problem of multivariate system optimization. The characteristics of the problem necessitate: 1) black box optimization; 2) possible application to nonconvex response surfaces because the system behaviour is unknown a priori; and 3) minimizing iterations for reaching optima due to the metaphysics and multiscale modelling required, which increases computational time.

Statistical approaches for optimizing a large number of design parameters, such as worst case and Monte Carlo evaluations, have been proposed in a number of studies. Because of the large number of simulation instances and significant processing cost for these methodologies (DOEs), others have proposed ways to limit the number of simulations, including design of experiments (DOEs). On the other hand, the DOE technique (Sohel,2016) has a number of limitations, including the need for minimal parameter interactions and a maximum of three layers. Additionally, since these techniques are used at discrete places in the problem space, they may cause quantization issues during optimization.

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Other global optimization strategies may be used as well. Due to the large number of data sets and compute time for each data set, global optimization algorithms, for example, may need a significant amount of computing power. Global optimization techniques required between 272–2650 function counts to converge to the least value when applied to the "peaks" function in MATLAB, which is an example function for two variables. Machine learning using Bayesian optimization (BO) only needs 100 function counts to converge to the least when applied to the same function. The function count, which might be numerous for each iteration, indicates the number of objective function evaluations throughout the optimization process. Our study into machine learning approaches in this work was primarily motivated by this (Park, 2017). In addition, similar approaches might be used to solve nonconvex, black box optimization problems, which was a need.

The following conclusions are drawn from this chapter.

Because of the high heat load of electronic packaging, the electronics industry is always looking for ways to improve device performance. As a consequence of this need, high-performance cooling solutions were developed employing new technology and improved methods. Natural convection, single phase forced convection, and two-phase cooling systems with high heat transfer coefficients have all been used to cool electronic equipment throughout time. Single phase liquid cooling systems have gained a lot of interest in recent years due to their ability to remove large volumes of heat. Because of their extraordinarily high heat transfer coefficient, liquids are considered a viable option for cooling high-powered circuits and CPUs.

In the literature, there are three basic methods for researching conjugate natural convection: analytical, numerical, and experimental. Because the cooling system is complicated and contains multiple characteristics, the analytical technique is inadequate, and it is limited to extremely simplified situations (Bilir, 2009). Using a numerical method, many challenges connected with complexity during experimentation may be resolved.

Traditional heat transfers experimental approaches employ thermocouples to provide exact data, but they have severe limitations. Thermocouples, for example, can only measure temperature at distinct points on a surface, therefore they can't tell you about the temperature distribution (Gerardi et al., 2010). You will save time and effort if you use the suitable numerical method to handle the difficulties listed above. The border and commencement conditions, on the other hand, must be accurate for trustworthy outcome estimations. A thorough literature

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evaluation of single-phase liquid cooling systems was conducted as a prelude to this research project, and some important conclusions may be summarised as follows:

1. The majority of the heat transport investigations inside the enclosures were done in a steady state and in two dimensions.
2. An increase in the Rayleigh number improves thermal performance.
3. The maximum temperature was influenced by the position of the hot and cold walls.
4. The heat transmission is heavily influenced by the substrate's thermal conductivity.
5. In most prior investigations, one or more walls were considered to be isothermal. The performance of the heated surface structure can be improved by employing parallel plate fins for instance.
6. The maximum temperature for the computer CPU should not exceed 85°C. The bulk of the investigations were conducted in a dimensionless format, with temperature measurements not given in absolute terms.

The majority of researchers that studied temperature distribution inside electronic chips and fluid flow in a single phase arrangement within a closed enclosure assumed that one or more parts of the enclosure walls acted as an isothermal heat sink. To employ this condition, force air or water against a cold wall, with the flow rate controlled over time to maintain the wall isothermal. Moreover, despite the fact that electronic components are employed in transient operating circumstances throughout their lifespan, steady state numerical studies account for more than 90% of the numerical studies for electronic equipment, according to Parry et al (2001).

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3. NUMERICAL ANALYSIS FOR THERMAL OPTIMISATION

Establishing the governing equations of the issue to be addressed is the first step in any numerical simulation. The 3-D governing equations for fluid mechanics and heat transport, as well as the basic turbulence models, are explained and provided in this chapter. There is also a brief consideration of the boundary conditions. The fluid mechanics equations are created using the universal laws of conservation of momentum, mass, and energy. The mass conservation equation is used to get the continuity equation, while the momentum principles are used to obtain the motion equations. The conservation of energy is represented by the fundamental law of thermodynamics. A continuity equation, three momentum equations for each direction of motion, the energy/heat equation, and a state pressure relation fill out the formulation of the Navier-Stokes equations.

3.1 Governing Equations

The governing equations for steady, three-dimensional, compressible, viscous flow are presented in this section.

3.1.1 Continuity Equation

The continuity equation is created using the law of mass conservation, which is based on the continuity principle. According to this theory, the difference between the rates of input into and efflux out of a space must match the rate of development of fluid mass contained inside that space. According to Eulerian, the differential version of mass conservation is as follows:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \tag{3.1}$$

where is the fluid density and u,, w are the velocity components in the x, y, and z

The term in first explains the build-up, which the second term must counteract. The convective term, which is the remainder of the equation, reflects the in-out mass flow from the boundaries, directions, respectively. The continuity equation is simplified to when the density of an incompressible fluid does not vary.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \tag{3.2}$$

where is the fluid density and u,, w are the velocity components in the x, y, and z

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3.1.2 Momentum as flow

Newton's second law of motion is applied to a fluid moving through an infinitesimal fixed control volume to yield the momentum equations (known as the momentum principle). The momentum notion provides a link between a flowing fluid's velocity, pressure, and density. When the fluid is assumed incompressible, the momentum equation has the following generic form.

$$\nabla.\sigma + f = \rho\left(\frac{\partial V}{\partial t} + V.\nabla\vec{V}\right) \tag{3.3}$$

The stress divergence is the first component, which is made up of the pressure gradient and viscosity$(\mu\nabla^2 V - \nabla p)$.The second term is body forces, whereas the right term is inertia. After numerous phases and combining various equations, the Navier-Stokes equations for incompressible fluid are formed

3.1.3 Energy Equation

The final version of the energy equation in three dimensions when the first law of thermodynamics is applied to a small control volume for an incompressible fluid is: "Energy entering the element + energy generated within the element = change in internal energy + energy out of the element." This may be quantitatively stated as follows:

$$\rho c\left(\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} + w\frac{\partial T}{\partial z}\right) = K\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) + q + \varphi \tag{3.4}$$

where q: Heat generation

c: Specific heat

φ: Viscous dissipation which could be written as

CFD may be used to analyse the Navier-Stokes equations shown above. CFD is effectively used in a wide variety of applications. For example, hydraulics, engine flows, aerodynamics, hydrodynamics, and so on. The spectrum of applications is vast, encompassing a wide range of fluid phenomena. Heat transport and fluid flow are examined using computer simulation methods in this research. There is a variety of available CFD software that may be used to analyse such situations using computers.

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3.1.4 Turbulence Models

In turbulent flow, the velocity, temperature, and pressure alter due to the uneven movement of the fluid's particles. As a result, time-averaged data and fluctuations may be used to illustrate turbulent flows. The most difficult job in CFD modelling is resolving the turbulence impact. The Reynolds Number, which is the ratio of inertial forces to viscous forces, determines whether the flow is unstable in a turbulent condition. The random variations of the velocity component and the pressure term make exact analysis difficult, if not impossible, in turbulent flow, even when numerical techniques are used. The quantities may easily be broken down into mean or time averaged values, as well as variable components.

It is extremely difficult to solve turbulent flow numerically using Navier-Stokes equations. This needs very fine resolution for time where the computing time becomes considerably infeasible for calculation. Because of the capacity to study a "stable" component and establish simpler deterministic connections for the fluctuating part, the Navier-Stokes equation may be solved with modest computer effort.

There are several turbulence models that employ time averaging. The most common models include $k-\varepsilon$ models, $k-\omega$, Reynolds Stress models, Spalart-Allmaras models, models, and so forth. None of them can produce satisfactory results for all types of flows, and each has advantages and downside

- The $\boldsymbol{k-\varepsilon}$ Model

The two equation turbulence models technique, often known as the two equation turbulence models, is frequently used in industrial CFD applications. This model is often used in turbulence models in a variety of variants, including STD k-epsilon, RNG k-epsilon, and realisable k-epsilon. Although the k model is easy to solve and has a quick convergence time, it does not perform well when pressure gradients are considerable. This approach is classified as two equations since the transport equation is solved for two turbulent variables, k and epsilon

There are many areas where $k-\varepsilon$ model is used for turbulence models to solve the Navier-Stokes equation numerically.

- Standard $\boldsymbol{k-\varepsilon}$ model:

  The standard $\boldsymbol{k-\varepsilon}$ model is extensively used in industrial CFD flow and heat transfer applications because it is accurate and inexpensive for a wide range of turbulent flows.

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The standard $\boldsymbol{k-\varepsilon}$ model, on the other hand, is only applicable for completely turbulent flows.

- RNG $\boldsymbol{k-\varepsilon}$ model:

  The RNG k -epsilon model is similar to the traditional k model, but it contains an additional term in the equation to account for the interplay between turbulence dissipation and mean shear stress, which improves accuracy for fast strained flows. The RNG k model improves predictions for high streamline curvature and strain rate, as well as Transitional flows.

- The realizable $\boldsymbol{k-\varepsilon}$ model

  In comparison to conventional $k-\varepsilon$, this model includes a novel formulation for turbulent viscosity. The word realisable refers to the model's ability to satisfy specific mathematical limitations on Reynolds stresses that are compatible with turbulent flow physics. The realizable $k-\varepsilon$ model improved performance for flows including rotation, recirculation and strong streamline curvature. It provides the best performance of all the $k-\varepsilon$ model versions for complex flows.

- Reynolds Stress Model (RSM):

  The RSM turbulent model is more stringent than previous models in accounting for the impacts of streamline curvature, swirl, rotation, and large variations in strain rate. For complicated flows, the RSM model provides reliable forecasts.

3.2 Boundary and Initial Conditions

Boundary conditions are required for the equations of motion on both sides of the domain in which the solution is sought, as well as on all surfaces of any objects inside the domain. Proper starting and boundary conditions must be used while solving the Navier-Stokes equation and the continuity equation. Boundary conditions are an important aspect of the Navier-Stokes equation solution in heat transport and fluid mechanics. The value of the variables at the border should be known for each issue to be addressed. The correctness of the findings is inextricably tied to the precision of the boundary depiction. There are many different models for turbulent flow that may be used to solve a problem, but some of them need more effort on the boundary conditions to match the model's requirements, making it more difficult. To arrive at the answer with the least amount of computing time, a fair evaluation of the different parameter values, such as velocities and temperature, is required for the beginning circumstances.

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3.3 Parametric Modelling of Heat Sink

The lumped system approach will be used to simplify the mathematical model in this section. The lumped model as a whole will be discussed, as well as three approximations: heat convection dominated, thermal radiation dominated, and an approximate mixed convection-radiation model.

Lumped Heat Sink Model

Consider a heat sink that was installed in a temperature-controlled environment. The power absorbed by the heat sink is dispersed via heat convection and thermal radiation heat transfer rules. The energy balance equation is as follows:

$$\frac{d(\rho cvT)}{dt} = p - hA(T - T_a) - \sigma A(T^4 - T_a^4) \qquad (3.5)$$

And the heat transmission area A is calculated as follows,

The given equation is nonlinear with regard to temperature and may be solved using the fourth-order Runge-Kutta technique (RK4). Further simplifications allow approximate analytical solutions to forecast the temperature of the heat sink as a function of time in the subsections that follow.

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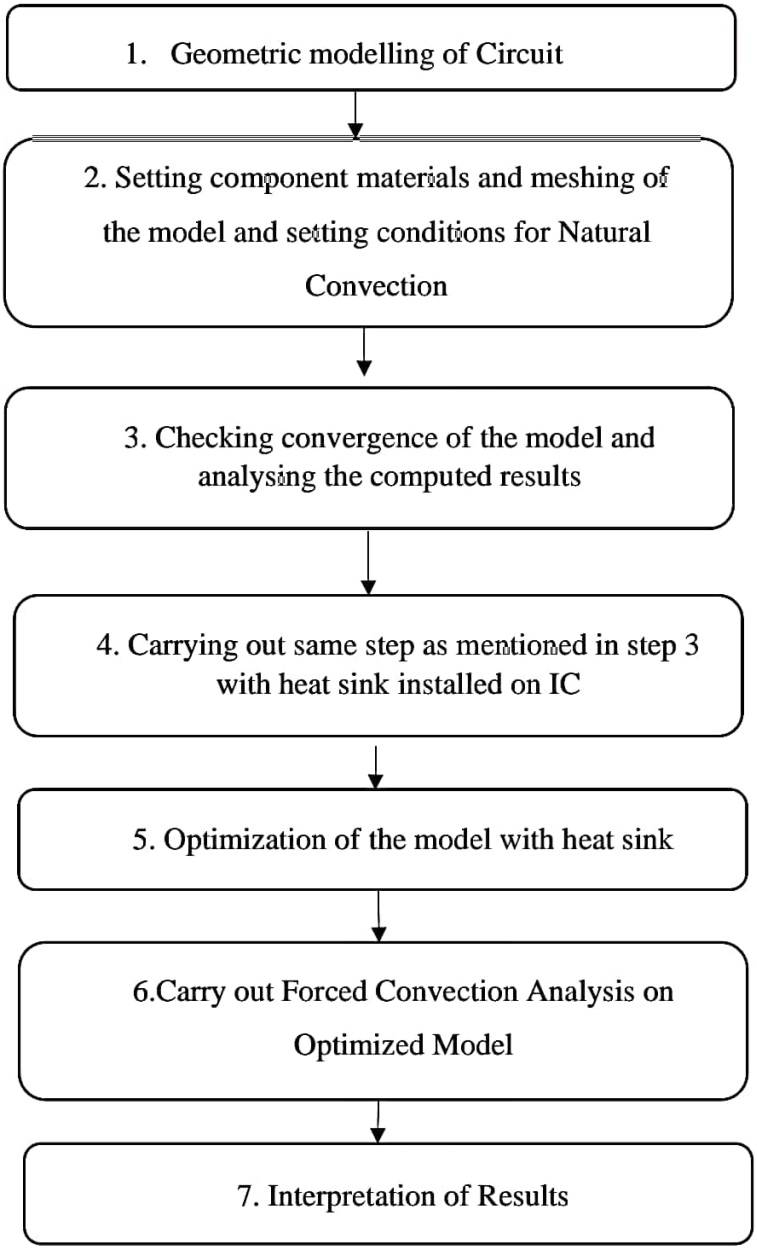
4. METHODOLOGY

For carrying out thesis research analysis, a model circuit was fabricated in Ansys Icepack software. The below figure represents the methodology adopted for this thesis. After the geometric modelling of circuit was done, setting of material of component was carried out. The selection of material is crucial since the accuracy of whole simulation depends on the material selected.



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After the selection of material was done, the meshing of the geometric model was carried out. In this simulation process, I carried out a multi-level meshing strategy which is a feature in Icepack software. In this study level of 0 – 4 meshing level was selected, 4 being the finer mesh whose mesh size is $1/4^{th}$ the size of mesh of level 0. The more the finer the mesh parameters, the better is the convergence of the model. After that natural convection parameters are set for the model.

In the third stage of the model convergence analysis of the model without heat sink is carried out. This step is necessary as for getting accurate results of the model. If the result obtained is divergent in case either the finer meshing is to be carried out or material selected for the component is not accurate.

In the fourth stage, the main objective is to minimize the overall distribution of the temperature along the board is carried out. For this purpose, the main source of heat in the model is identified mitigation procedure is carried out. In this case the ic that controller the motor signals would be the main heating element of the circuit. To mitigate, this overheating of this IC, heat sink was installed in the model. Firstly, user input values input into to the model and result was analysed and

In the fifth stage, with the help of Ansys parameter optimization of the heat sink model was carried for getting optimum shape which decreases the overall temperature of the model. After setting parameters like width of the fin, height and gap optimization of the model was carried out.

In the final stage of the study, forced convection analysis of the model was carried out with th e implementation of heat sink in the model. In this case the flow regime considered as turbulent and turbulent model selected as k – epsilon model.

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4.1 Geometric Modelling of Circuit

The below figure represents the overall simplified architecture version of typical motor controller

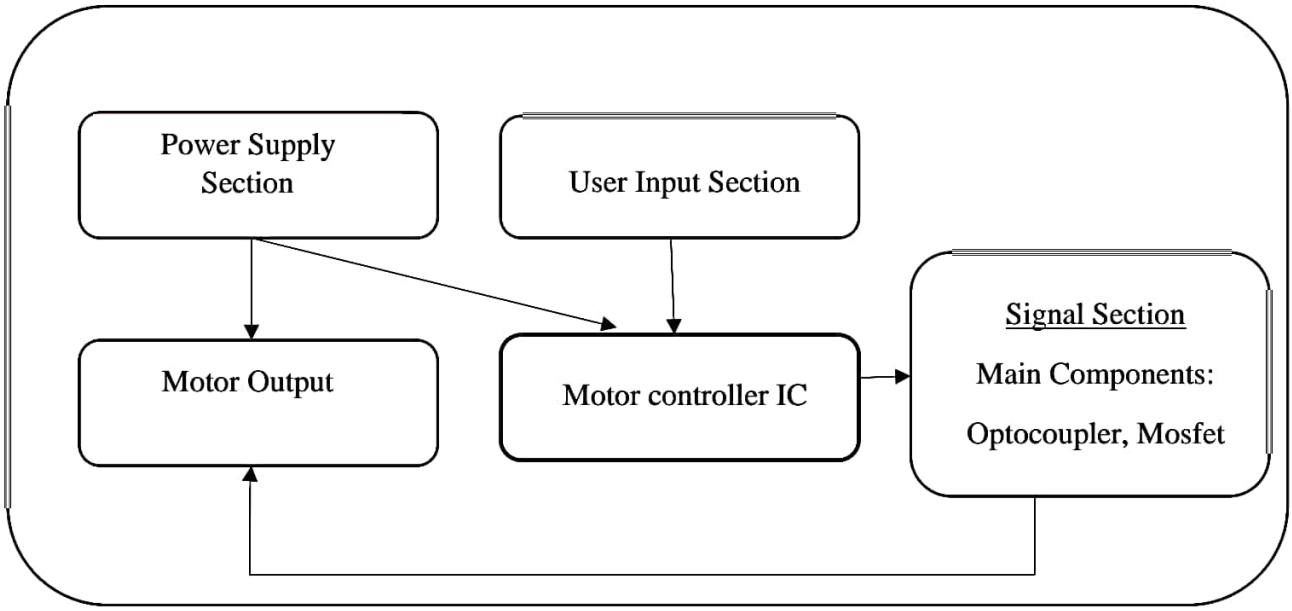

Figure 3.1 Motor Controller Architecture (created by author)

As shown in figure, the motor controller ic and power regulator ic's are the main areas of high temperature in the overall circuit. The overall working working of circuit starts from user input which is basically accelerator cable is fed to the motor controller ic via input sensors like potentiometer. The motor controller IC then computes the required signal and generates respective pulse width modulation signals (PWM) which is transmitted to opto-coupler and mosfets . This signal then reaches Motor output section.

The dimensions of electronic components in geometric model of the circuit was fabricated according to the specifications sizes that is available in the market. With reference to the typical motor controller circuit, geometric model of the circuit was modelled in Ansys Icepack.

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- **The modelled geometry in Ansys Icepak is given below**

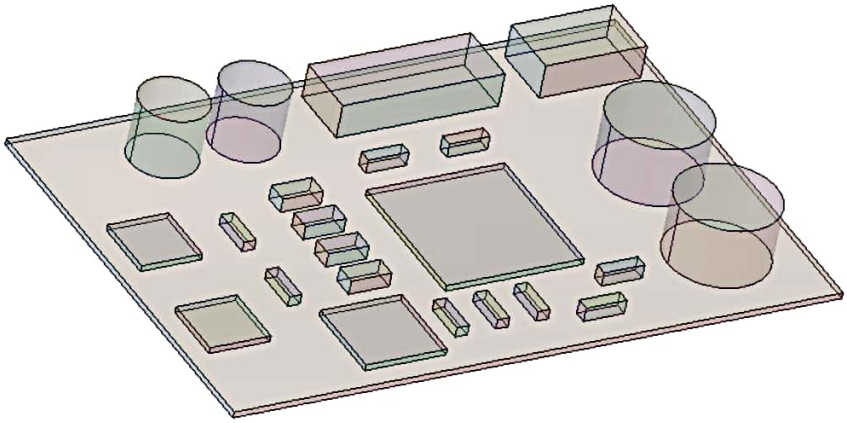

Figure 3.2 Isometric view of geometric model (created by author)

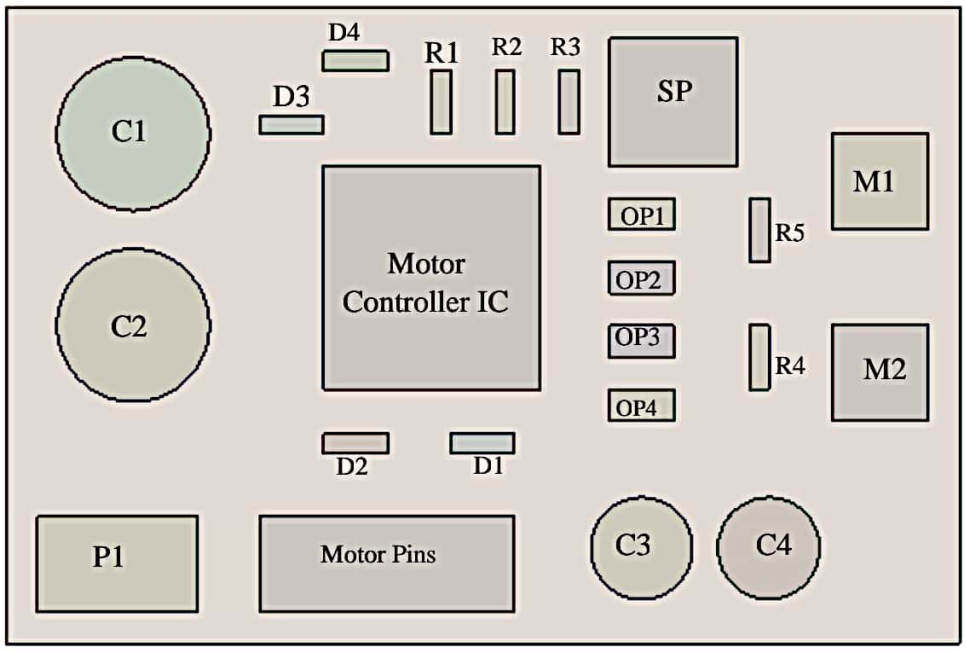

Figure 3.3 Top view of geometric model (created by author)

Notation	Component
C	Capacitor
P	Power Supply Pins
OP	Opto coupler
D	Diode
R	Resistor
SP	Support Ic
M	Mosfets

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4.2 Setting of The Simulation Environment for The Model

4.2.1 Setting of Material and Heat Output

In this phase material selection of electronic component are carried out for the simulation. In this simulation the motor controller ic is the primary heating source and is set with an input value of 20W, electrolytic capacitor and switching mosfets are set at a heat output of 1W and 2W respectively.

Table 3.1

The below listed components are used for the simulation (Created by Author)

Item	Material	Quantity	Heat Output Considered (W)
Motor Controller IC	Silicon	1	20
Electrolytic Capacitor	Ceramic	4	1
Diode	Silicon	4	0
Opto coupler	Silicon	4	0
Mosfet	Silicon	2	2
Resistor	Ceramic	5	0
PCB Board	FR4	1	0

4.2.2 Carrying out Meshing of the Model

For carrying out meshing process, in this study used multi-level meshing strategy in with a level of 1 – 4. The level 4 having the mesh size $1/4^{th}$ of that of the meshing size of the level 1 mesh element size.

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Table 3.2

The Meshing levels for each component (Created by Author)

Meshing Level	Component
Level 1	PCB Board
Level 2	Power supply and Motor Pins
Level 3	Controller ,Support Ic and Mosfets
Level 4	Resistor, diodes and opto coupler

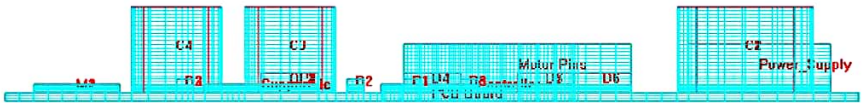

Figure 3.4 Side view of model after meshing

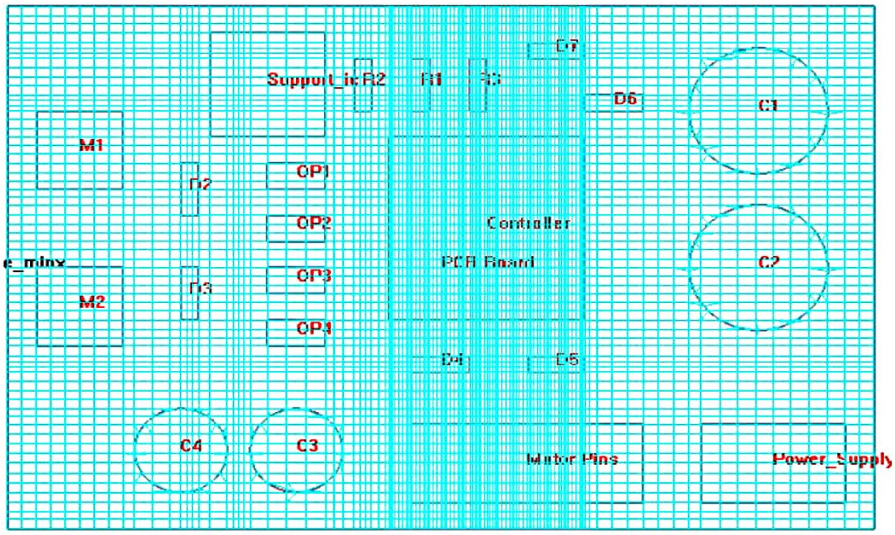

Figure 3.5 Top view of model after meshing

Table 3.3 Meshing Parameters (Created by Author)

Mesh Type	Mesher HD (Ansys Icepak)
Total number of elements	644499
Total number of elements	666626

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4.2.3 Setting of Heat Transfer mode

In this case, Natural convection heat transfer condition is considered since in most of the motor controllers don't use fans for cooling for achieving optimum power efficiency hence condition for forced convection is neglected. In this simulation, the entire board in enclosed in cabinet enclosure with opening is to the horizontal.

4.3 Setting of Solution Conditions

The simulation was carried out with the below mentioned settings. The solution would run for 200 iterations with parallel processing enabled. This conditions were uniformly across three conditions that is, model without heat sink, with heat sink model and lastly with optimized heat sink model.

Table 4.4 Simulation Parameters (Created by Author)

Condition	Setting
Turbulent Mode	On
Turbulent equation	K –epsilon SST Model
Mode	Natural Convection
Natural Convection Model	Boussinesq approximation
Operating Pressure	101325 N/m^2
Ambient Temperature	25°C
Radiation Condition	Off
Convergence Criteria	
Number of iterations	200
Flow	0.001
Energy	1e-7
Joule Heating	1e-7

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4.4 Modelling of Heat Sink

In this phase modelling of heat sink for the motor controller board was carried out. The initial dimensions of the heat sink were user defined. The below table show the parameters set for the modelling of heat sink in this simulation.

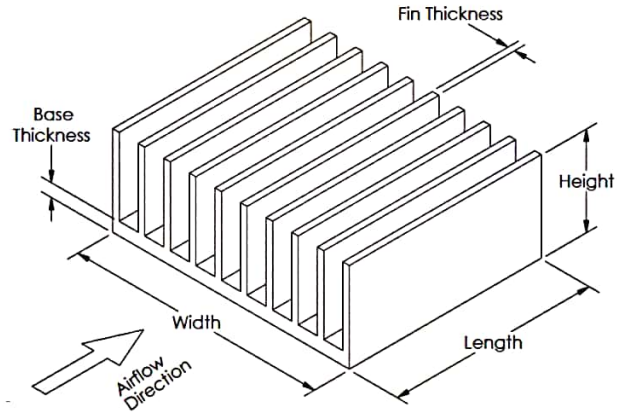

Figure 3.6 Heat sink dimensions

The above figure depicts the basic geometric parameters considered for modelling the heat sink. For this simulation the base dimensions of heat sink were considered same as that of topology dimensions of motor controller ic. The geometric parameters for the heat sink model was user defined. The parameters like the number of fins, the fin thickness of heat sink and base height of heat sink was considered as user defined parameters in this simulation. The dimensions of heat sink parameters were chosen according to the base geometry of controller ic chip. The below table mentions the parameters considered for the simulation of the user defined heat sink model for the modelled geometry.

Table 4.5 Initial Heat Sink Parameters (Created by Author)

Parameter	Size
Number of fins	8mm
Thickness of fin	2mm
Base height of heat sink	1.5mm
Overall height of heat sink	11mm
Material of heat sink	Aluminium

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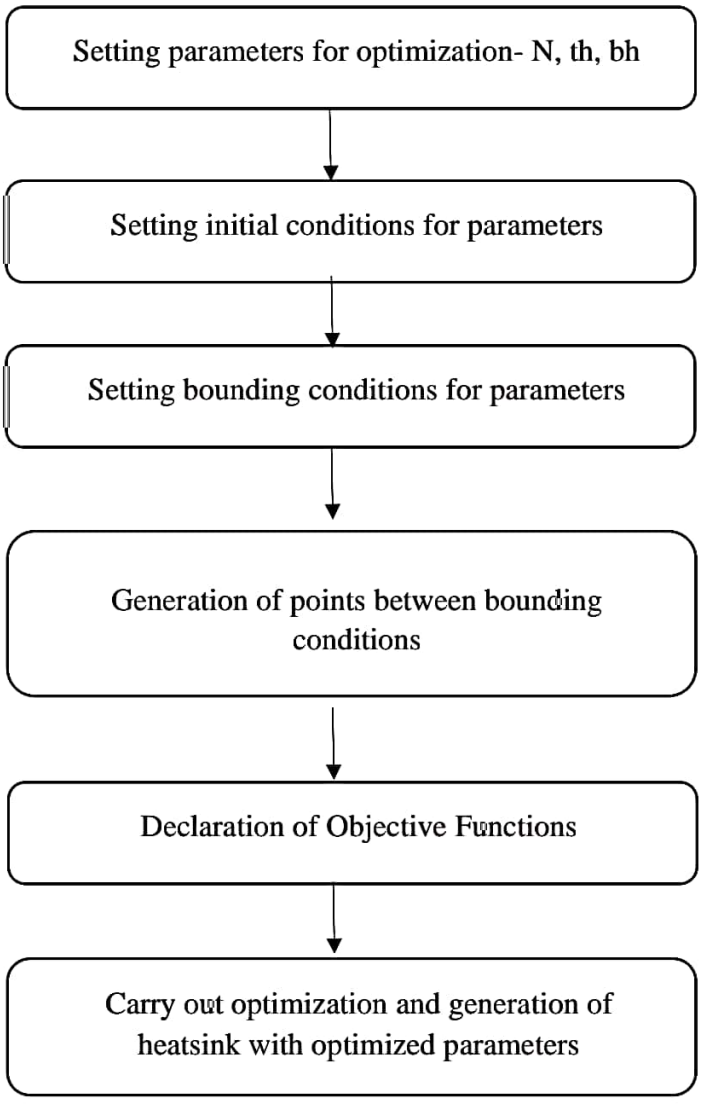
4.5 Carryout Heat Sink Optimization

In this phase optimization procedure was carried out for the user defined heat sink model. The main aim of the optimization routine is to minimize the overall temperature of the heat sink model. The below figure depicts the optimization routine carried out for this simulation



For carrying out optimization process, the above mentioned steps are being carried out. The main parameters for optimization such as Number of fins(N), thickness of fin (th) and base height of fin (bh) are values considered for optimization. For each points the a total of 200 iterations in convergence section were run to obtain values for the model at that particular point. Finally point which meets the objective function are selected and model is generated with that parameter.

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4.5.1 Setting of Parameters for Optimization

In this phase the parameters to be optimized are set. For this simulation, the parameters like base height (bh), number of fins (N) and thickness of fin (th) is considered as optimization parameters keeping the overall height of heat sink as user defined in order to reduce the computational cost

.

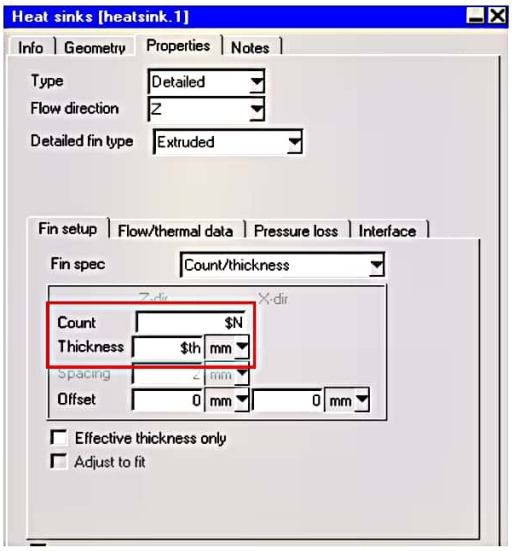

Figure 4.7 Heat sink Properties menu

As from figure parameters was declared by adding a variable preceded by $ sign instead of numerical values. Here in this case fin count '$N', thickness of fin 'th' and base height 'bh' in geometry menu are considered as parameters for optimization.

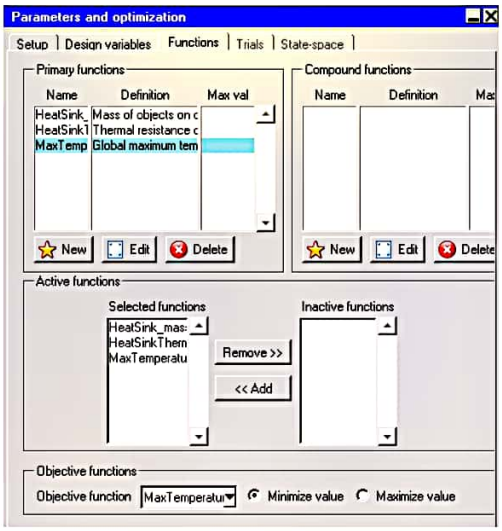

Figure 4.8 Parameter Optimization menu

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As mentioned in the parametric optimization menu parameters along with objective functions are declared. For this case, there are three objective functions are selected for minimization. They are 1. Maximum temperature of heat sink, 2. Mass of heat sink and 3. Thermal resistance of heat sink.

After setting the parameters for optimization the transfer of parameters along with objective functions to Ansys Workbench was carried out for further executing the optimization procedure.

In this phase the initialization of values for heat sink models like fin number, thickness of fin and base width of heat sink was carried out. For number of fins was initially set as 8, with fin thickness of 2mm, base width of heat sink model as 1.5mm.

For carrying out optimization procedure 'Response Surface Optimization' tool in Ansys workbench was linked with our icepack model having user defined fin dimensions.

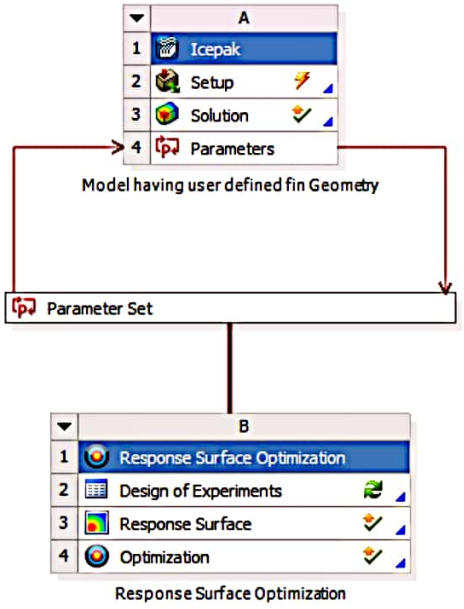

Figure 4.9 Workbench view of optimization routine for this simulation

4.5.2 Setting of Constraints for the Parameters

In this phase, after model initialization process for parameters was carried out, limits are set. Based on this limit, software generates random points for carrying out optimization process. The below table mentions the constraints set for this model.

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Table 4.5 Setting parametric constrains for the model (Created by Author)

Parameter	Constrains
Number of fins (N)	8-14
Thickness of fin (th)	Lower Bound: 0.5 mm Upper Bound : 2 mm
Base width of fin (bh)	Lower Bound: 0.5 mm Upper Bound : 2 mm

The constraints are set in 'Design of Experiments section' in Response Surface Toolbox in Ansys. The below figure shows the overall setup for this simulation model.

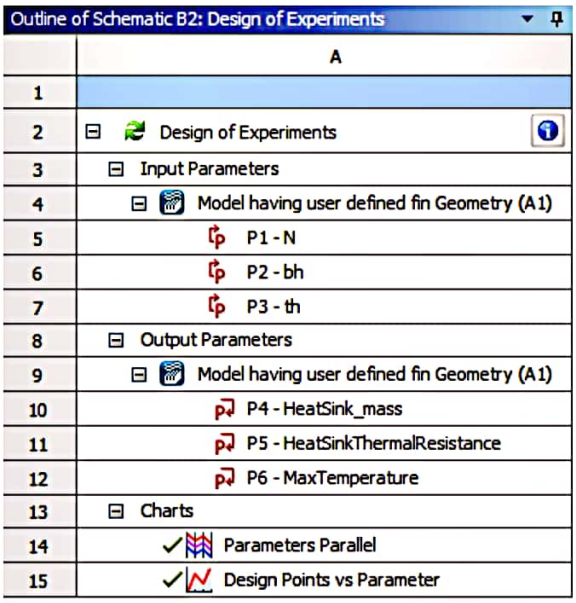

Figure 4.10 Overall layout of input parameters and objective functions

Based on this setup, the software generates random points and optimization routine was carried out this points with main aim of function to reduce the global temperature of the board.

- **Data points generated for optimization**

  The below figure depicts the data points generate for carrying out optimization for this model.

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1	Name	P1 - N	P2 - bh	P3 - th
2	1	8	1.25	1.25
3	2	8	0.5	0.5
4	3	8	2	0.5
5	4	8	0.5	2
6	5	8	2	2
7	6	9	1.25	1.25
8	7	9	0.5	0.5
9	8	9	2	0.5
10	9	9	0.5	2
11	10	9	2	2
12	11	10	1.25	1.25
13	12	10	0.5	0.5
14	13	10	2	0.5
15	14	10	0.5	2
16	15	10	2	2
17	16	11	1.25	1.25
18	17	11	0.5	0.5
19	18	11	2	0.5
20	19	11	0.5	2
21	20	11	2	2
22	21	12	1.25	1.25
23	22	12	0.5	0.5
24	23	12	2	0.5
25	24	12	0.5	2
26	25	12	2	2
27	26	13	1.25	1.25
28	27	13	0.5	0.5
29	28	13	2	0.5
30	29	13	0.5	2
31	30	13	2	2
32	31	14	1.25	1.25
33	32	14	0.5	0.5
34	33	14	2	0.5
35	34	14	0.5	2
36	35	14	2	2

Figure 4.11 View of data points for this model

With the above-mentioned points optimization procedure was carried out for the model. After optimization final optimized model was generated in Ansys Icepack. After optimization of heat sink was done, static analysis of the modelled geometry was carried out.

4.6 Carry out Forced Convection Analysis on Optimized Model

In this phase the optimized model geometry was undergone a forced convection analysis. There are several methods for carrying out this method. Some of them are: considering the entire model in an enclosure with vents / grills on the side faces, by supplying a constant velocity at one side. Another way of doing this by implementation of fan in the model. The fan should be place in a location in which the flow of air is parallel to the surface of the fins. The flow of particles in the simulation model was considered.

In this phase, the flow considered here is turbulent, which is different from the above mentioned three cases. The turbulent equation considered in this case is k-epsilon model. The below figure depicts the figure of fan curve which shows pressure and flow rate that the fan will produce for any system connected to it. The speed of fan is set at 1000rpm.

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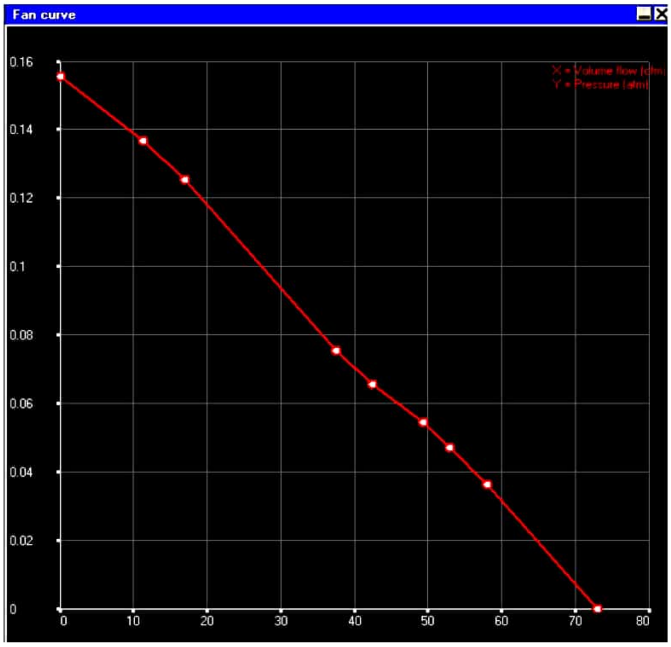

Figure 4.12 Fan Curve of fan used in the model

The below figure depicts the placement of fan and grill for the modelled geometry. The placement of fan is set parallel to the direction of fins.

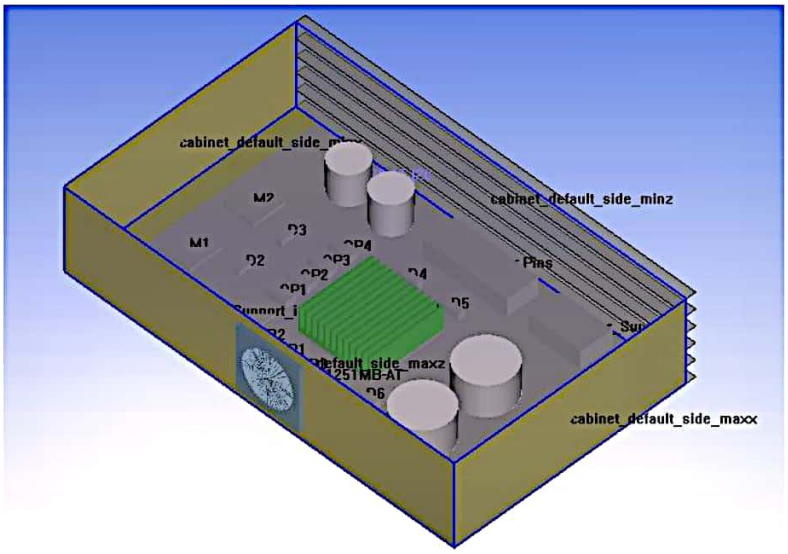

Figure 4.13 Isometric view of model with fan

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5. RESULT ANALYSIS AND INTERPRETATION

5.1 Evaluation of Convergence of the Model

The convergence plot of the geometric model describes whether the model is accurate enough for carrying out analysis of the model. In this simulation after numerous tweaking of meshing parameters and conditions, the below figure depicts the convergence plot for the model.

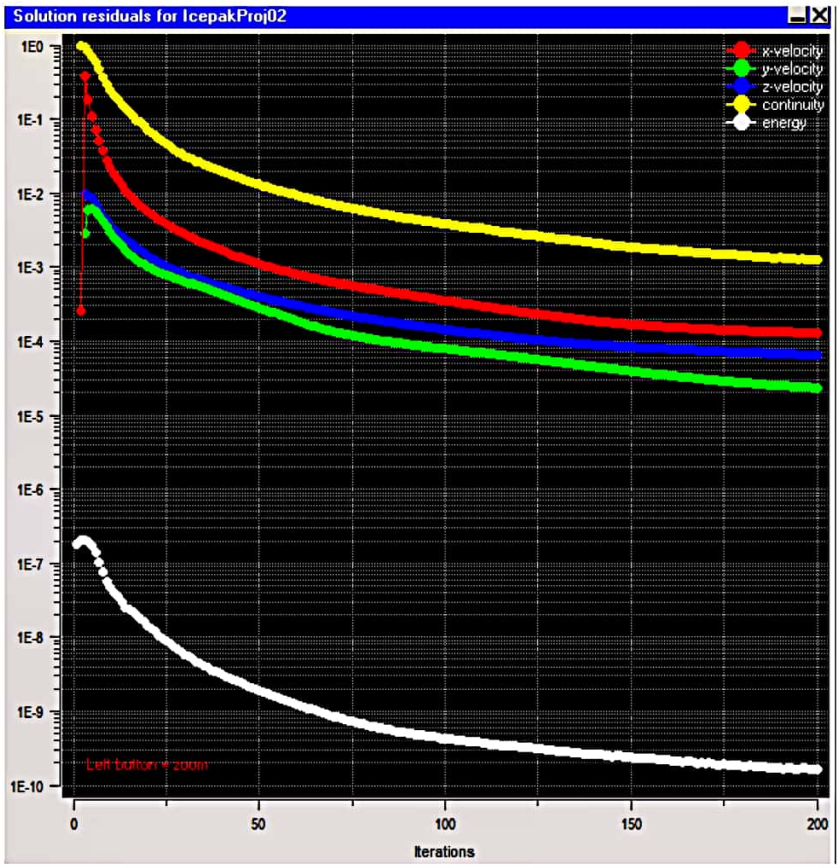

Figure 5.1 Convergence Plot for the modelled geometry

After running a total 200 iterations, the model has converged after 175th iteration of the simulation, which gives the verdict that the model is fit for carrying out further analysis. In the result analysis of the model primary aim is to find the distribution of temperature along the board and to find the maximum global temperature of the model.

Since the model has converged after analysing the result for geometric model without similar, routine of steps was carried out in case of model with user defined heat sink model and model with optimised heat sink.

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5.2 Evaluation of Model

The evaluation of the model was carried out in 4 stages,

1. Case 1: Temperature distribution of model without heatsink.
2. Case 2: Temperature distribution of model with heat sink
3. Case 3: Temperature distribution of model with optimized heat sink
4. Case 4: Temperature distribution of model under forced convection.

In the first case, the temperature distribution of the model without the application of heat sink was carried out. Critically analysed the areas of maximum heat dissipation and the flow of heat in the circuit board. Based on the evaluation critical heat spots will be analysed, and as mitigation method, would first carry out a method to increase the heat distribution of the board. In the second stage based on the results obtained from the Case 1, study of the modelled geometry, a user defined model of heat sink would be generated. The base dimension of heat sink would be equal to the maximum heat source of the board. The same evaluation procedure was carried out as mentioned in the first case. With comparing the results with previous case, would analyse if the application of heat sink was effective in the modelled geometry. If the model performance increased with application of heat sink, then optimisation procedure for getting a heat sink model would be carried out, else the different method would be carried out.

In the third stage, optimisation of the heat sink model was carried out. The parameters for optimization considered for this simulation are Total number of fins (N), base width of heat sink (bh) and thickness of fin (fh) was considered as optimization parameters. The optimisation of heat sink was carried out by the input of initial values for parameters. After that software generates random points between the range specified for each parametric values. With this points model computes the global temperature values at each points. Then after simulating model with parameters specified, final parameter was selected by the software and software automatically generates the optimized geometry for the model. In the last stage, the temperature distribution of model was analysed with the implementation of fan in the model. Finally, results were compared and analysed in each stage of simulation.

5.2.1 Case 1: Evaluation of Model Without Heat Sink

In this case the geometry modelled was evaluated without the application of heat sink. After convergence criteria is met for the model temperature distribution across the model and maximum temperature of model was analysed.

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- **Analysing The Temperature Distribution of the Model**

The below table depicts the final observed temperature across model after the simulation procedure was completed. The components of temperature above 120oC considered as critical components indicated as red, were thermal optimization need to be carried out

Table 5.1

Temperature Distribution of components without application of heat sink (Created by Author)

Component		Temperature (°C)
Capacitor	C1	43.7
	C2	28.31
	C3	25.4
	C4	25.1
Motor Controller		154.3
Diodes	D1	32.8
	D2	26.26
	D3	43.65
	D4	43.7
	D5	86.04
	D6	97.01
Mosfets	M1	34.5
	M2	25.61
Motor Pins		26.17
Opto coupler	OP1	45.52
	OP2	39.43
	OP3	35.61
	OP4	31.05
PCB Board		154.3
Power Supply		25.25
Resistors	R1	108.51
	R2	77.67
	R3	113.07
Support IC		63.29

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In this section, main focus is to evaluate the temperature distribution of the modelled geometry

The figure 5.2 and 5.3 depicts the top view and side view of modelled geometry respectively. As from figure it is clear that without the application of heat sink the model temperature reaches up to 154°C.From figure the temperature of controller is the maximum and supporting capacitors temperature is minimum at 25°C. Hence for extending the life of modelled geometry, mitigation methods to be carried out to reduce the overheating controller ic.

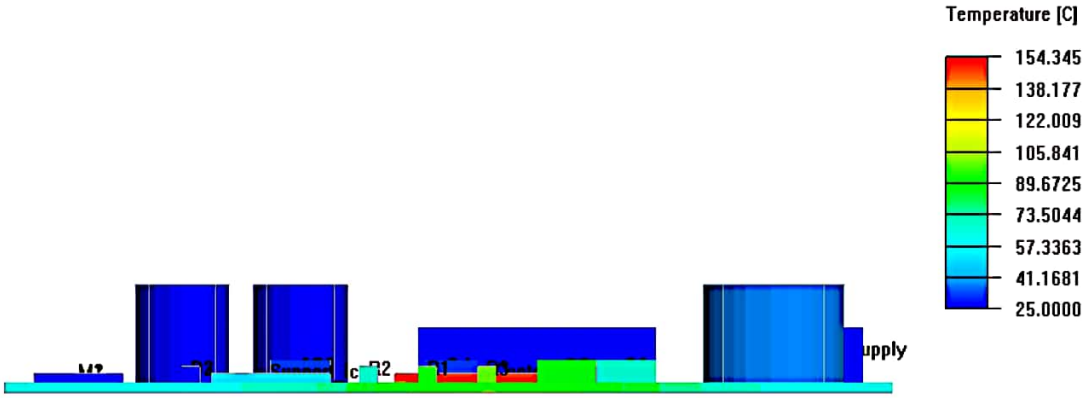

Figure 5.2 Side View of the temperature distribution of model

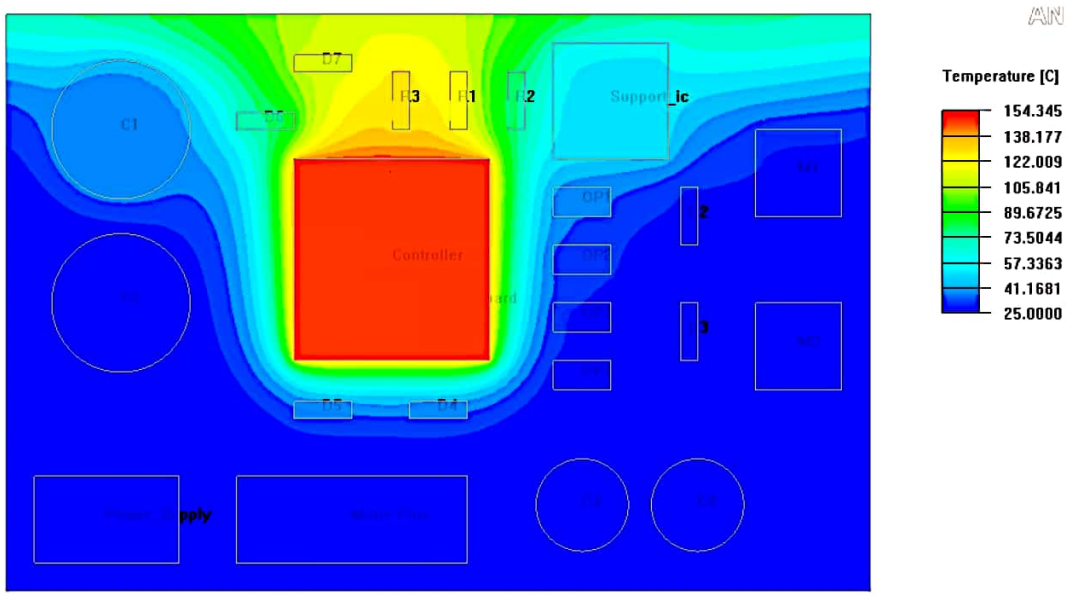

Figure 5.3 Top View of the temperature distribution of model

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The figure 5.4 depicts the isometric view of model. As the condition is set as natural convection with opening as considered with inlet and outlet as mentioned in the below figure with gravitational acceleration applied perpendicular to the board. The flow regime formed after applying these conditions is mentioned in below figure.

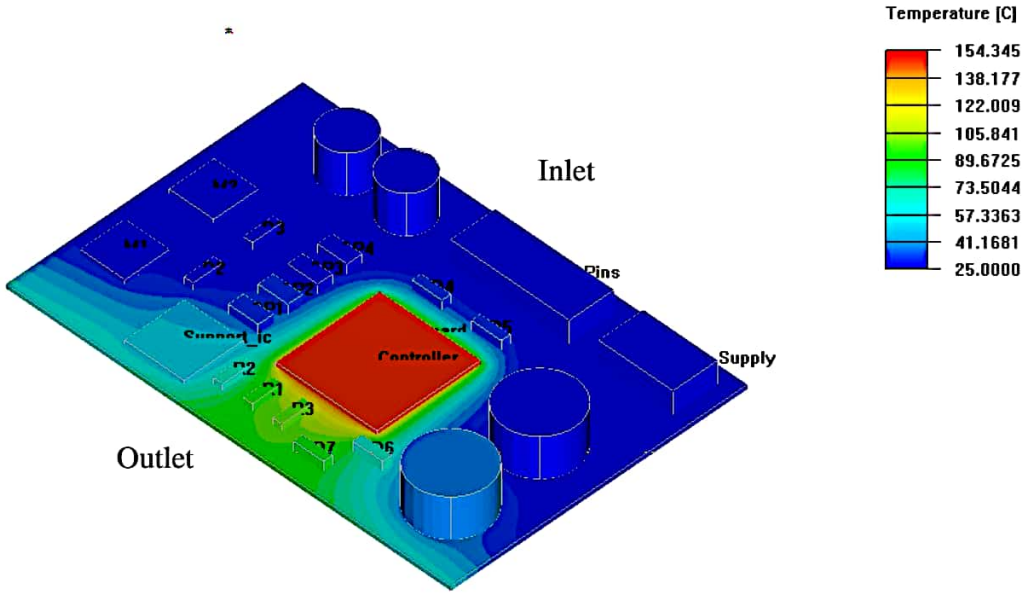

Figure 5.4 Isometric view of the temperature distribution of model

Pressure Distribution of the Model

The below figure 5.5 depicts the overall pressure distribution of model obtained after simulation. The portion near the resistor's experiences higher pressure than rest of components in the board

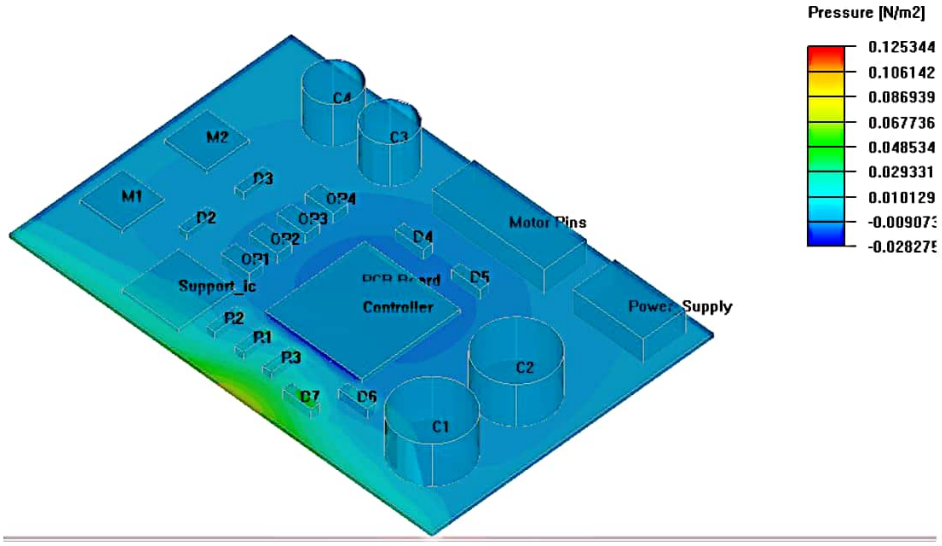

Figure 5.5 Isometric view of the pressure distribution of model

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5.2.2 Case 2: Evaluation of Model with Heat Sink

On analysis of model in first case, it was clear that the thermal mitigation method should be carried out for our modelled geometry. In this case, the application of heat sink was adapted as a mitigation method for controlling the temperature distribution of the model. The dimensions of heat sink were user specified, with base geometry is equal to the same dimension of controller ic. After this same procedure was carried out as mentioned that of case 1.

Analysing The Temperature Distribution of the Model

Table 4.2

Temperature Distribution of components with application of heat sink (Created by Author)

Component		Temperature (°C)
Capacitor	C1	41.2
	C2	26.2
	C3	24.8
	C4	25.2
Motor Controller		126.7
Diodes	D1	31.5
	D2	25.3
	D3	42.4
	D4	42.7
	D5	84.6
	D6	94.3
Mosfets	M1	32.6
	M2	23.8
Motor Pins		25.4
Opto coupler	OP1	44.6
	OP2	38.9
	OP3	34.62
	OP4	31.1
PCB Board		126.1
Power Supply		24.27
Resistors	R1	107.35
	R2	75.89
	R3	111.7
Support IC		62.9

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The below figures 4.6 represents the temperature distribution of model with heatsink with user defined parameters. As from the figure, from viewing in the mentioned direction, the areas near heat sink, that is to the right of heat sink near power supply, the temperature is high as compared to the areas were the switching circuits, that is mosfets and voltage support capacitor section for controlling the operations of support ic.

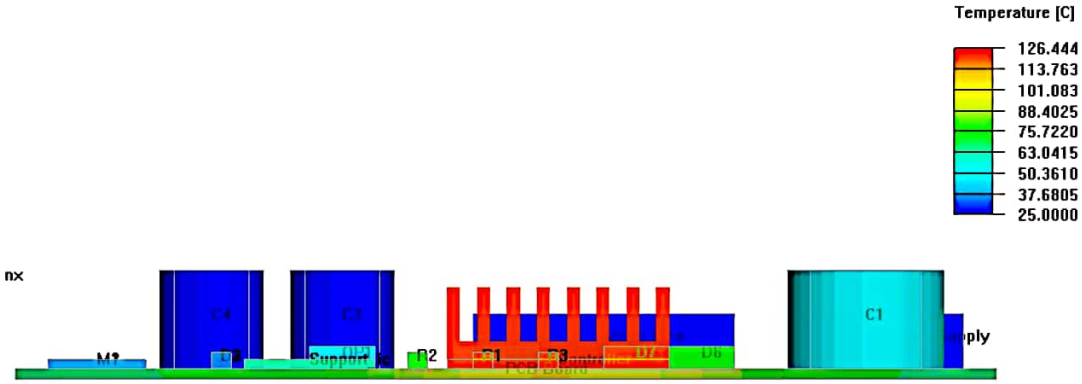

Figure 5.6 Side view of the Temperature distribution of model with heat sink

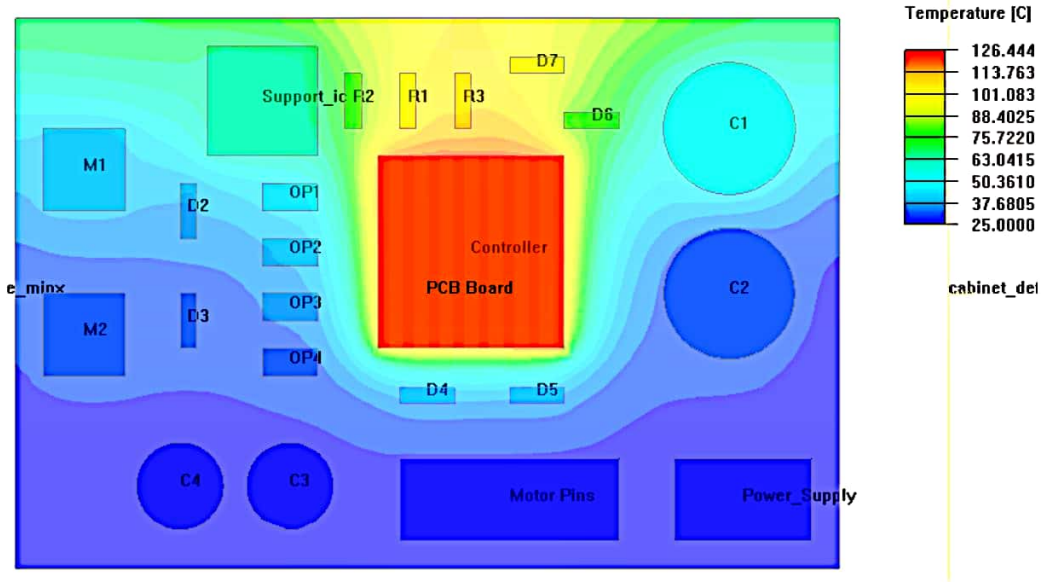

Figure 5.7 Top view of the Temperature distribution of model with heat sink

The above-mentioned figure 5.7 depicts the top and side of the modelled geometry. As from figure it is clear that with the application of heat sink the model temperature reaches up to 126°C in comparison to 154°C with model without heatsink, hence the application of heat sink to the modelled geometry was effective. Since the modelling of heat sink was based on user defined inputs, there is certainly a room for improvement.

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The below figure 5.8 temperature distribution contour obtained when model is simulated for natural convection condition. In this phase no external velocity is applied and flow is considered as laminar. From the isometric point of view, it is clear that the distribution of high temperature is more towards the resistor section of the model.

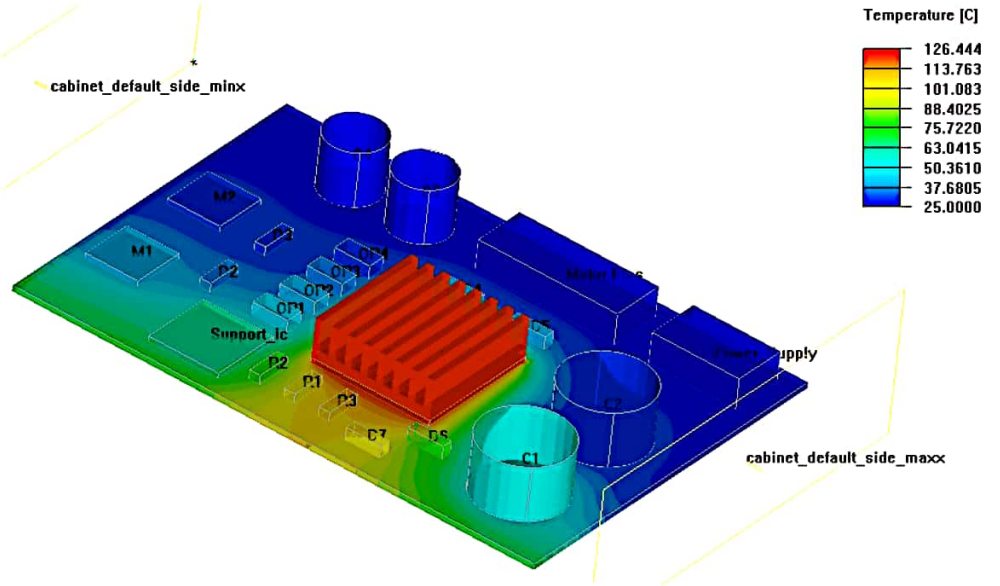

Figure 5.8 Isometric view of the Temperature distribution of model with heat sink

The below figure 5.9 depicts the pressure distribution of the model with heat sink. From isometric point of view of model, it indicated that the distribution of pressure is maximum at areas near the resistor section close to the heat sink. On simulation of model, the model achieves a maximum pressure value of 0.09 N/m2 at areas near the heat sink of the model.

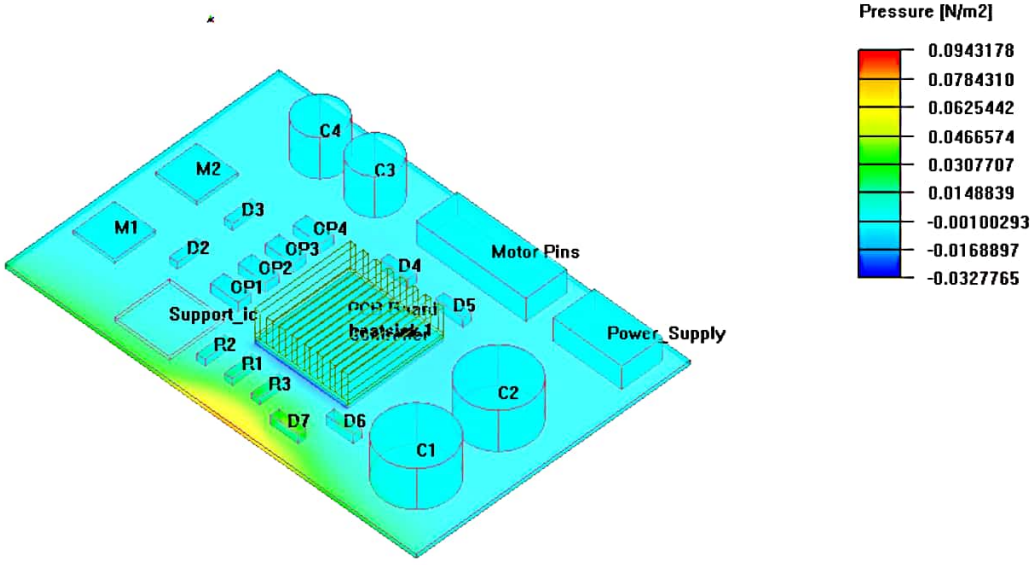

Figure 5.9 Isometric view of the Pressure distribution of model with heat sink

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5.2.3 Case 3: Evaluation of Model with Optimized Heat Sink

In this stage the optimization of heat sink model was carried out, with main aim to reduce the overall temperature distribution of the model. For carrying out this operation the parametric optimization in Ansys was utilized making the number of fins (N), the base height of the heat sink (bh) and thickness of fin (th) as the parameter to be optimized. Then after the computation of model at different points, the final optimized results is represented as below.

Analysing The Temperature Distribution of the Model

Table 5.3

Temperature Distribution of components with application of optimized heat sink (Created by Author)

Component		Temperature (°C)
Capacitor	C1	44.02
	C2	28.4
	C3	25.4
	C4	25.12
Motor Controller		108.71
Diodes	D1	33.32
	D2	26.38
	D3	39.57
	D4	38.79
	D5	72.38
	D6	88.34
Mosfets	M1	35.19
	M2	25.70
Motor Pins		26.18
Opto coupler	OP1	44.47
	OP2	38.25
	OP3	34.40
	OP4	30.34
PCB Board		108.6
Power Supply		25.28
Resistors	R1	92.68
	R2	71.58
	R3	96.6
Support IC		61.52

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The below figure 4.10 depicts the side view of model obtained after simulating model with optimized heat sink. From figure it is clear that the distribution of maximum temperature of 108°C is more at the centre portion of the board.

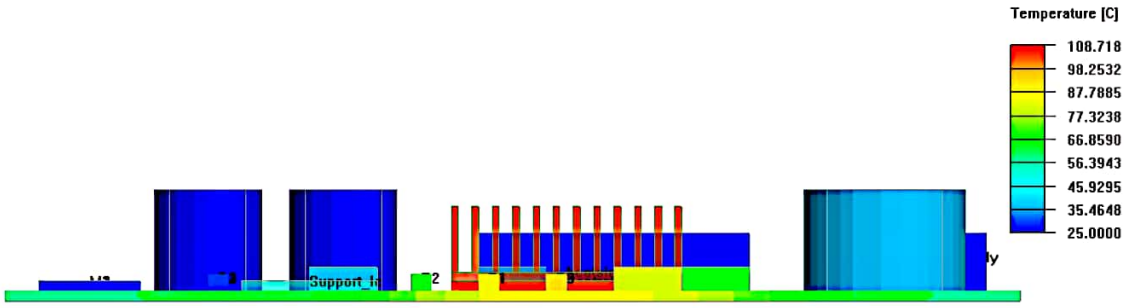

Figure 5.10 Side view of the Temperature distribution of model with Optimized heat sink

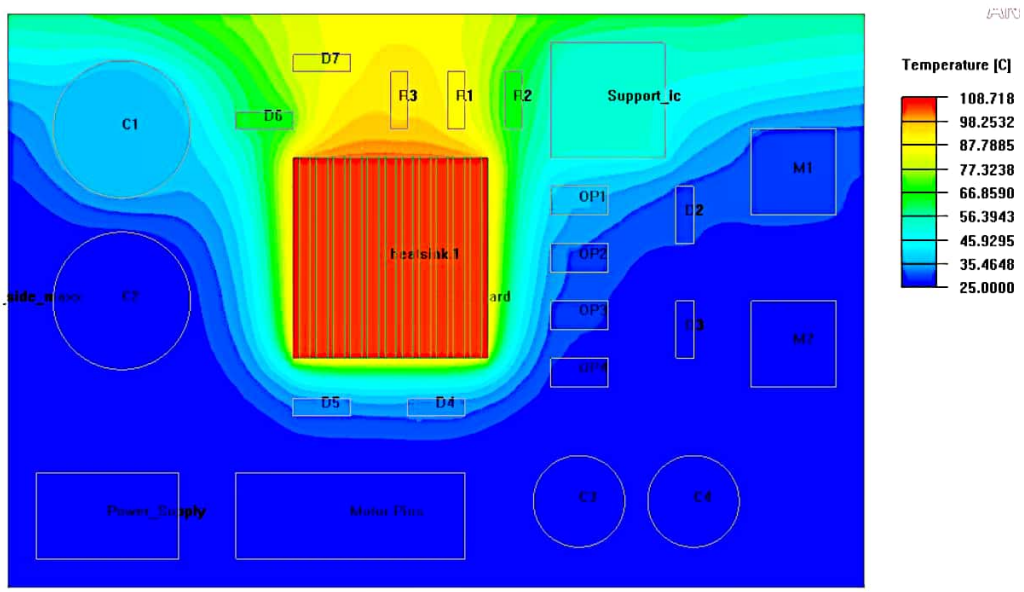

Figure 5.11 Top view of the Temperature distribution of model with Optimized heat sink

The figure 5.11 depicts the top view of the model with optimized heat sink. It is evident from this figure that heat flow is more towards outlet section of model. Since the model is simulated with opening on both sides one considered as inlet and other as outlet, without considering input of applied velocity The heat flow direction obtained by the model is depicted as mentioned in above figure.

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The below figure 5.12 depicts the isometric view of optimized heat sink model. After the simulation model achieved a maximum temperature of 108°C, which is very good results when comparing the results with distribution of heat in model without heat sink which is about 154°C and 126°C that for with user defined heat sink model. So thermal optimization of the modelled geometry or circuit board was achieved.

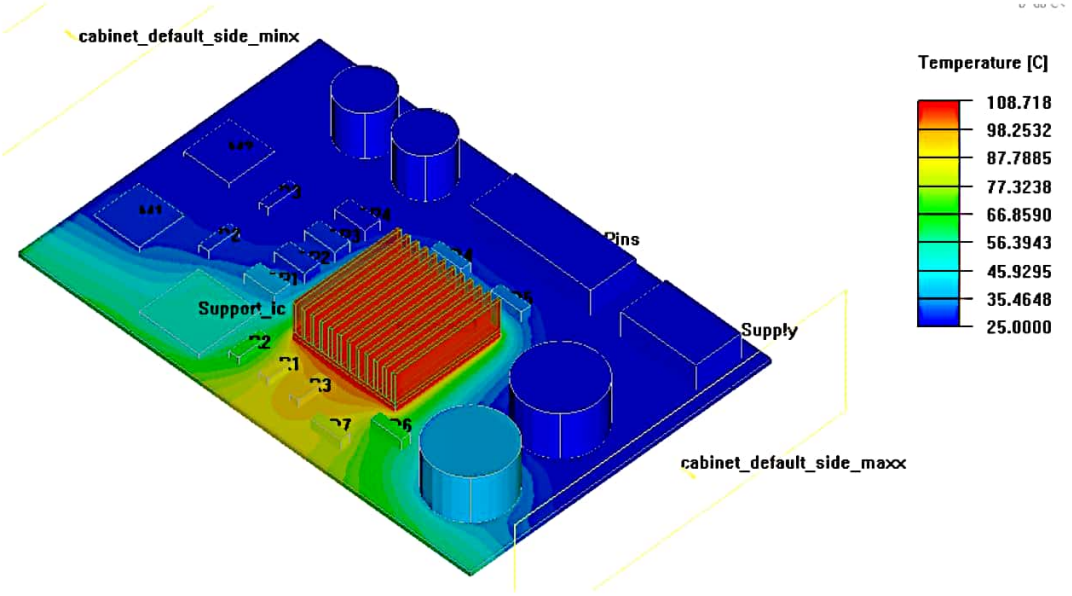

Figure 5.12 Isometric view of the Temperature distribution of model with Optimized heat sink

The figure 5.13 represents the pressure distribution of modelled geometry. After simulation, the model achieved a maximum pressure of 0.06N/m2 and a minimum pressure of 0.01N/m2 both located at top and bottom portion of heat sink.

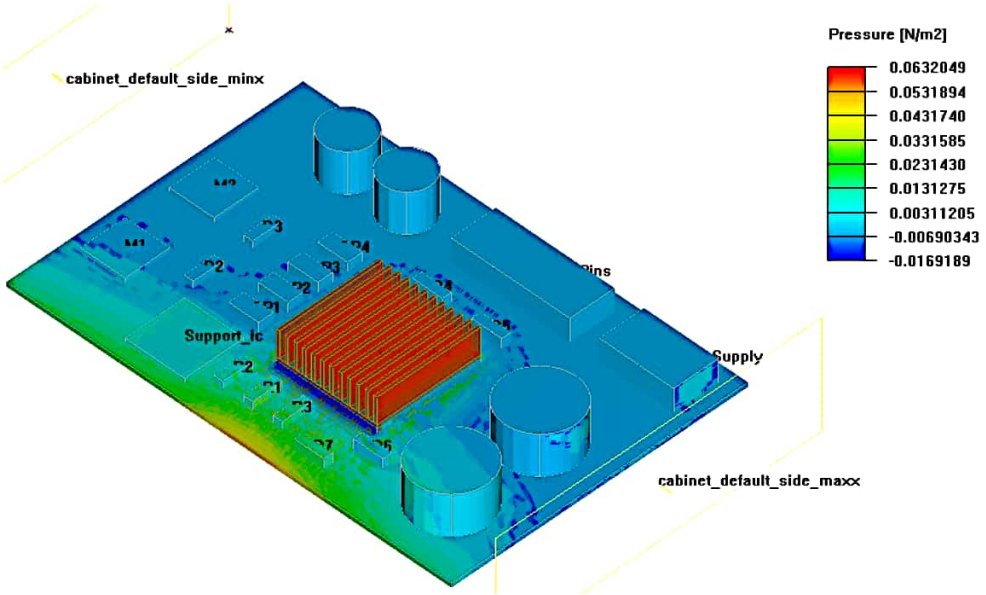

Figure 5.13 Isometric view of the Pressure distribution of model with Optimized heat sink

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5.2.4 Case 4: Evaluation of Model with Optimized Heat Sink with Implementation of Fan (Considering Forced Convection)

In this stage the thermal performance of model was evaluated under forced convection condition. The speed of fan was considered as 1000 rpm. The opening of enclosure geometry considered in above 3 cases was replaced by with fan opening on one side and grill on side that is opposite to the fan geometry. The simulation in this case was carried out by considering flow as turbulent were in other three cases the simulation environment considered as laminar. The turbulent model chosen was k-epsilon turbulent model.

Table 5.4

Temperature Distribution of components with application of optimized heat sink and fan (Created by Author)

Component		Temperature (°C)
Capacitor	C1	27.34
	C2	27.31
	C3	29.64
	C4	28.68
Motor Controller		53.9
Diodes	D1	26.82
	D2	27.78
	D3	31.65
	D4	31.64
	D5	26.64
	D6	27.2
Mosfets	M1	27.21
	M2	28.9
Motor Pins		29.43
Opto coupler	OP1	26.93
	OP2	27.45
	OP3	27.91
	OP4	28.52
PCB Board		53.75
Power Supply		29.10
Resistors	R1	26.29
	R2	26.36
	R3	26.15
Support IC		26.83

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The below figure5.14 depicts the side view of model obtained after simulating model with optimized heat sink with the application of fan whose direction is parallel to the direction of heat sink arrangement. From figure it is clear that the distribution of maximum temperature of 54°C is which is great improvement from a temperature of 108°C in case of optimized heat sink.

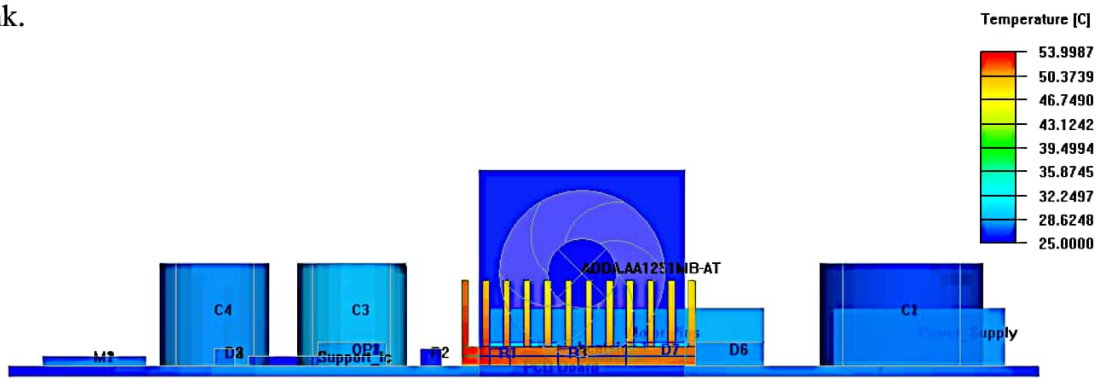

Figure 5.14 Side view of the Temperature distribution of model with Optimized heat sink and fan

The below figure 4.15 depicts the isometric view of the modelled geometry under forced convection condition. From figure, it is clear that area near the flow path of fan has lower temperature in comparison to exit outlet condition. The temperature around 36°C was obtained on the portion after heat sink.

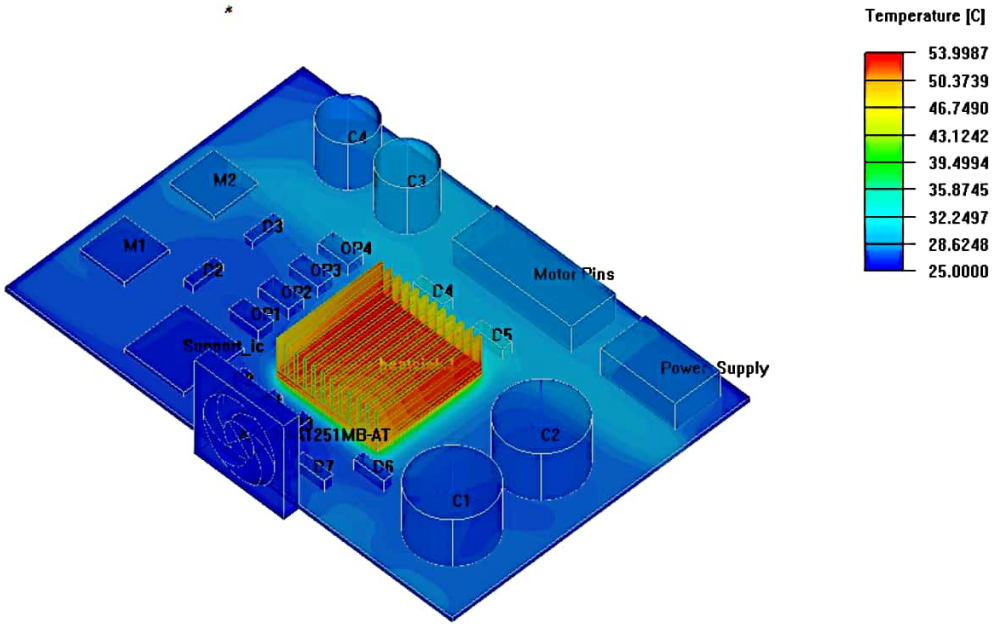

Figure 5.15 Isometric view of the Temperature distribution of model with Optimized heat sink and fan

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The below figure 5.16 depicts temperature of model from top view. As from figure, it is clear that portion of heatsink opposite to the fan has higher temperature distribution in the board.

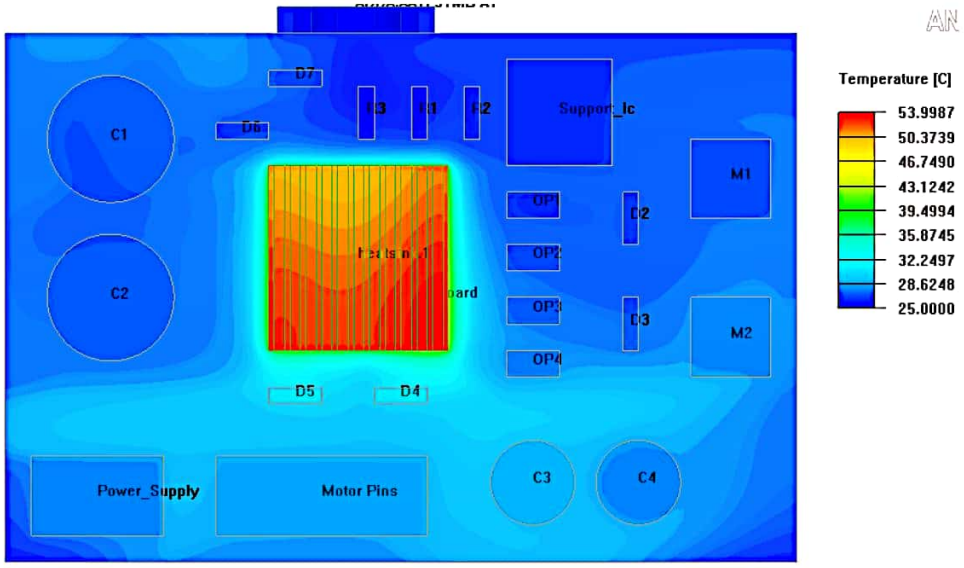

Figure 5.16 Side view of the Temperature distribution of model with Optimized heat sink and fan

- **Pressure Distribution of the Model**

The below figure 5.17 show the overall pressure distribution of the model. As from figure, it is clear that the maximum pressure of the model was evaluated the portion of heat sink close to the fan inlet area

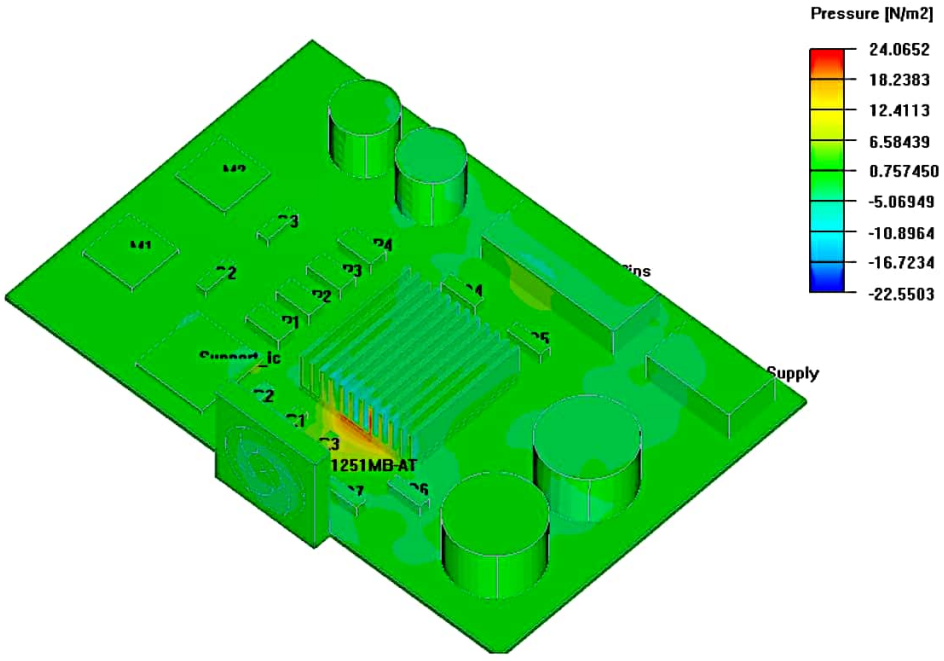

Figure 5.17 Isometric view of the Pressure distribution of model with Optimized heat sink and fan

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- **Velocity Distribution Contour of the Model**

In this section velocity distribution of the modelled geometry was depicted. The below figure 4.18 and 4.19 illustrates the distribution of velocity of air in the model. From fan the velocity of air slightly increases as it passes through the gaps of the fins. After passing through fin geometry, the path of air gets obstructed by the enclosure walls. Due to this a recirculation phenomenon occurs and splitting of air passage occurs which will result in slight increase in velocity in both ways.

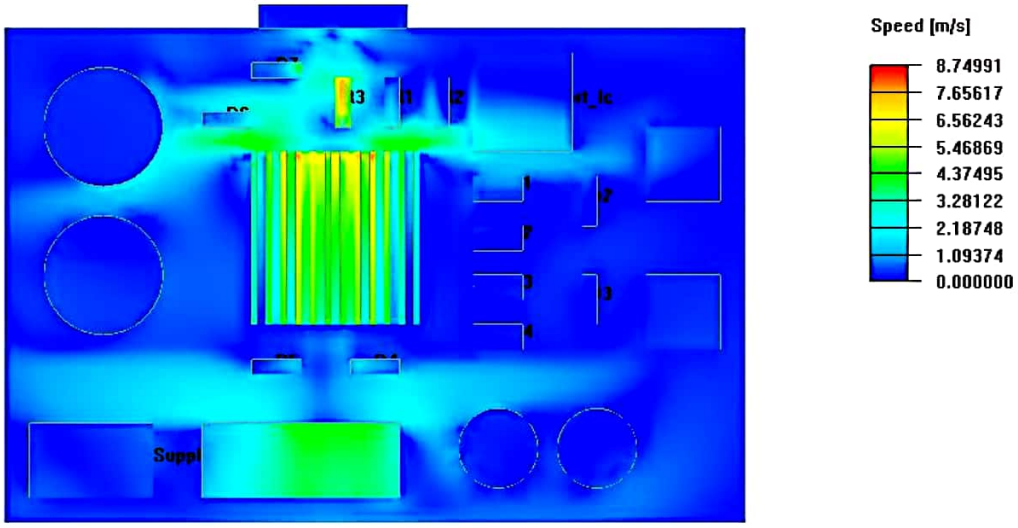

Figure 5.18 Top view of the velocity distribution of model with Optimized heat sink and fan

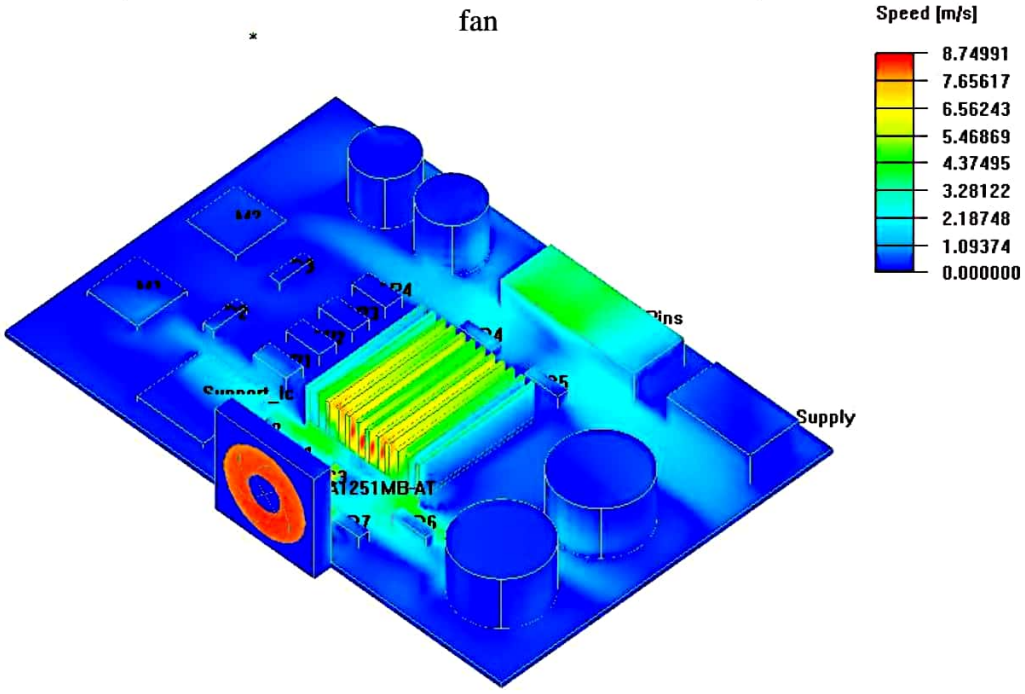

Figure 5.19 Isometric view of the velocity distribution of model with Optimized heat sink and fan

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- **Particle Trace Representation of the Model**

In this section, particle representation of the modelled geometry was carried out. Here in this case the flow path of air stream from fan was illustrated. As mentioned in the figure (Figure 4.20 and 4.21) the velocity stream of air for this simulation reaches at about 8.7m/s. Since fan was placed in an axis parallel to the gaps of fins, the flow from fan enters the fins and hits the walls of the enclosure and then recirculates throughout the board, hence reducing the temperature of board globally

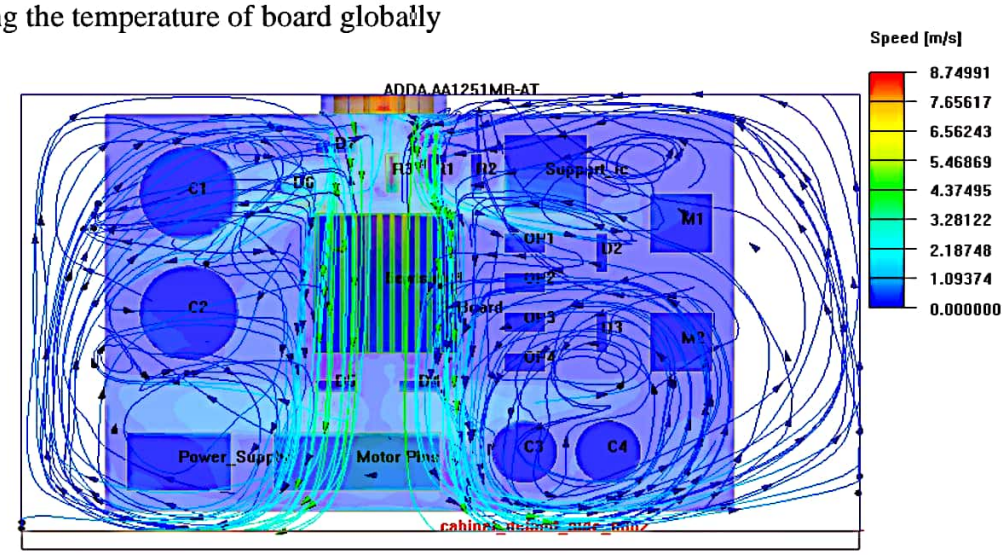

Figure 5.20 Top view of the Particle trace distribution of model with Optimized heat sink and fan

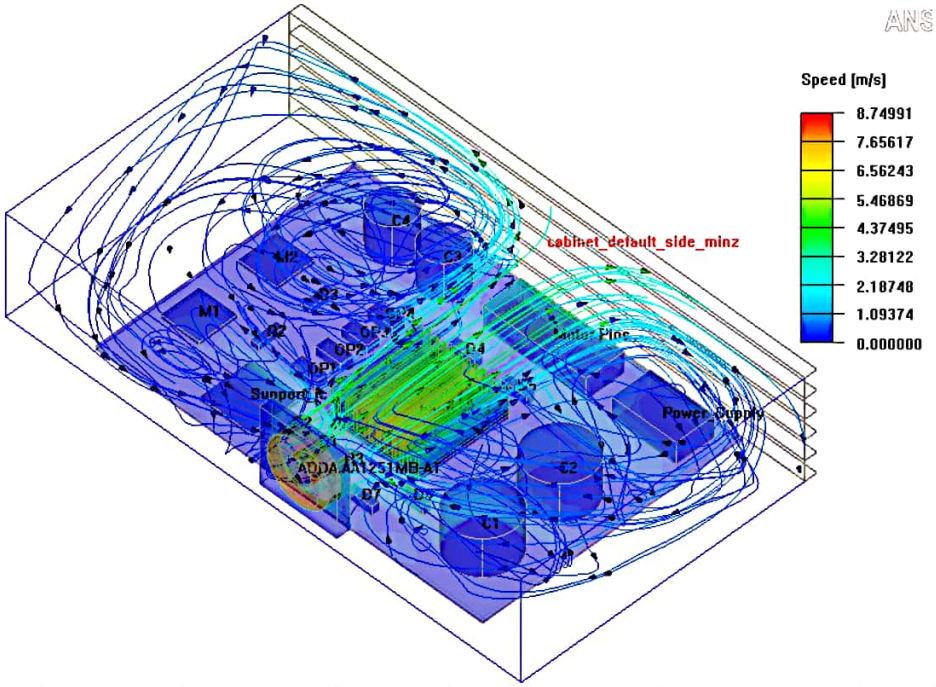

Figure 5.21 Isometric view of the Particle Trace distribution of model with Optimized heat sink and fan

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- **Velocity Vector Representation of the Model**

In this section velocity vector representation of the modelled geometry was represented (Figure 4.22 and 4.23). The below figure represents the top and isometric view of velocity representation of the modelled geometry. With fan velocity set at 1000rpm, the velocity of model at inlet was computed as 8.7m/s. When coming to the heat sink section the velocity was noted as 7.5 m/s. As from figures, it is clear that rest of velocity in the board other than heatsink and fan flow path is around 1- 2m/s.

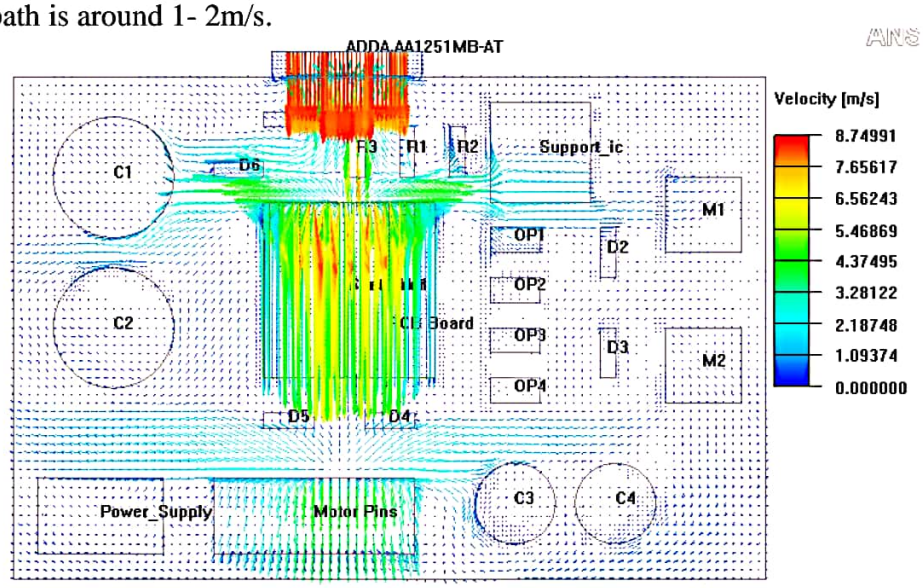

Figure 5.22 Top view of the velocity vector distribution of model with Optimized heat sink and fan

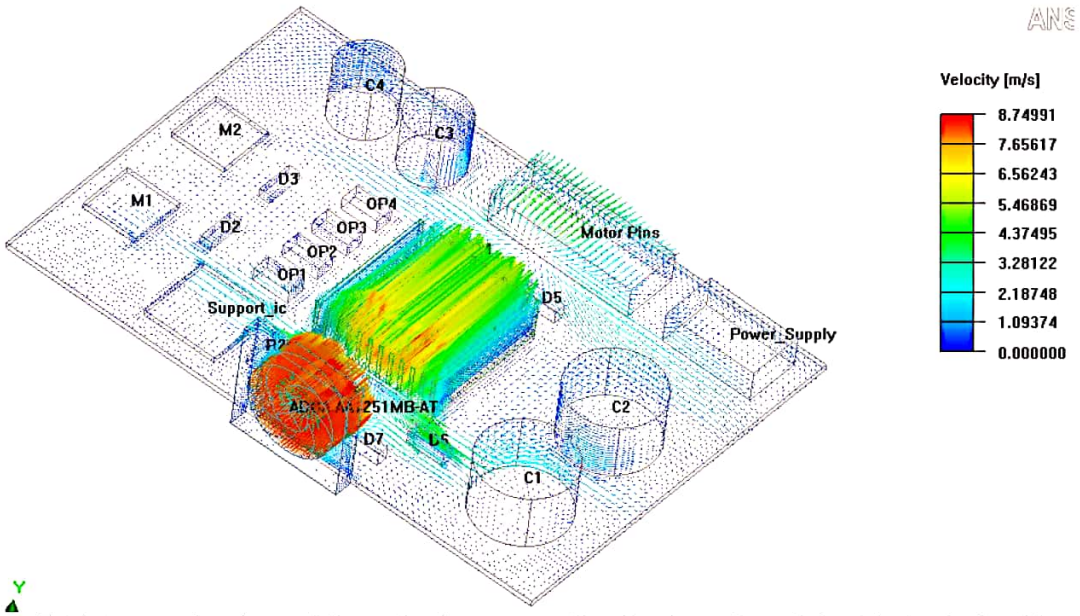

Figure 5.23 Isometric view of the velocity vector distribution of model with Optimized heat sink and fan

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5.3 Interpretation of Results

On analysis of thermal management of the model geometry it was found following results are obtained which is depicted in a table as mentioned below.

Table 5.5

Summary of Temperature Distribution of Model Obtained from four different cases (Created by author)

Case No:	Test Case	Motor Controller IC Temperature (°C)	Heat Flow Type	Fluid Flow Type
1	Without Heatsink	154.3	Natural Convection	Laminar
2	With Heat sink	126.7		
3	With Optimized Heat Sink	108.7		
4	With Optimized Heat Sink & implementation of fan	53.9	Forced Convection	Turbulent

The table 5.5 represents the results obtained by simulating the thermal distribution of board under 4 different cases. It is clear from table that thermal optimization of electronic circuit can be achieved by taking proper measures like placement of highly heat emitting source at centre of circuit to make the heat distributed all over the board. In this simulation, motor controller ic is considered as heat source, so the placement of motor controller ic was set at middle of board.

For the first case, the simulation of board was executed without the application of heat sink. The heat flow condition was considered as natural convection with fluid flow type as laminar. In this case the circuit was enclosed in a cabinet with two openings which simulates the environment for natural convection. The controller heat output considered was 30W which is same for all other cases. Here the influence of gravitational acceleration is considered. With setting the initial conditions for simulation such as laminar flow, steady state condition, setting total number of iterations to 1000 final obtained temperature at motor controller ic is found to

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be 154°C.From the temperature distribution contour, it is evident that reducing the temperature of controller ic is necessary as it affects the life expectancy of other neighbouring components.

For mitigating or reducing the temperature of controller IC, in the second stage implementation of heat sink was carried out. The base geometry of heat sink was considered as same as that of controller ic, the heat sink geometry was initially user defined. The number of fins considered as 8 and thickness of fin considered as 2mm.The same simulation environment was considered in this case as mentioned in the first case. The flow is considered as natural convection and fluid flow as laminar. After meshing operation, the simulation was carried out with 1000 iterations and increased if necessary to meet the convergence criteria. After simulating the model, the modelled board attains a temperature of 126.7°C, which is great improvement of temperature obtained from the first case. The temperature distribution plots and results shows that model temperature can be further reduced with optimization of heat sink.

In the third case, optimization procedure for achieving the thermal efficiency of modelled geometry was carried out. In this setting environment conditions are similar to that of previous cases. The heat transfer mode is selected as natural convection and fluid flow considered as laminar. For carrying out optimization routine, minimization of heat sink temperature is considered as objective function. The parameters like the number of fins, thickness of fin and base width of heat sink is considered as parameters for optimization. The optimization procedure was carried with 200 iterations for each point, optimized values are selected and model is generated based on the optimized values. After optimization procedure completed the model attained a temperature of 108°C which shows the effectiveness of optimization of heat sink.

In the fourth stage, a different environment set was considered. In this case the application of fan in the model was introduced to notice any significant change in result is observed. Due to the implementation of fan, the heat flow type considered in this case was forced convection. The fluid flow regime considered as turbulent. Since Ansys Icepack provides libraries for different fan models, one of fan model was selected and its geometry was modified in order to make fit to our modelled geometry. The placement of fan was set in direction that is parallel to the fin structure. And simulation was carried out with forced convection conditions for 1000 iterations. The results such as overall temperature of board, particle trace of model, velocity distribution of the model, pressure distribution of the model was carried out. The final obtained temperature was found to be 54°C which is very significant result as compared to the other three cases of the model.

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6. CONCLUSIONS

Thermal management in circuit designing is a crucial part which determines the overall life expectancy of circuits. As the operating temperature of electronics circuit increase the less is the probability of circuit to sustain for long time. Adding to this, increase in circuit operating temperature also affects the performance of components interlinked to it. So, to increase the reliability and durability of circuits, thermal management should consider a crucial process in circuit fabrication process.

- In this thesis, main aim is to design a model motor controller board, and analyse the temperature distribution of the board. The main reason to select motor controller board for analysis is that since the shift towards EV is growing day by day and the motor controller which is like heart of EV, since the controlling of motor and performance of ev motor depends on motor controller, thermal management of motor controller is necessary. In this simulation, a simplified form of motor controller board is fabricated in Ansys Icepack. The main components considered are motor controller ic, mosfets, capacitor, opto-coupler, diodes and resistors.
- In the first section of this thesis, a brief discussion of the need of thermal management in electronic circuit is discussed. From study it was understood that, in most cases the natural convection method is employed as a mitigation method for cooling electronic circuits. Adding to this placing the most heating ics in the middle of board, attaching the mosfets with small heat sinks, placing of fans on the top of mostly heating ics are some of the approaches taken by the integrated circuit industry for achieving the thermal efficiency in electronic circuits. Following this, different thermal management techniques employed in electronic circuits, discussed various related works that is carried in the field of thermal management, various optimization strategies are briefly mentioned in the literature section of this thesis. After evaluating all the related works in this field, it was clear that for efficient design of thermal management of circuit CFD modelling of the electronic circuit is necessary for achieving optimized thermal efficiency of circuit. The selection of materials, orientation of electronics also adds to this.
- In the second section of the thesis, a methodology was formulated. In this section a model circuit was fabricated. The dimensions of the circuit board was taken in reference to the market available component dimensions. The motor controller board was selected

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as selected as reference circuit board for carrying out analysis. A simplified version of the controller circuit selected for modelling. Here in this thesis ,H-bridge circuit which commonly used circuit for controlling motor is selected. The board consist of motor controller Ic which is the main heating source, two mosfets which controls the direction of the motor from signal input from motor controller ic, diodes and resistors for controlling the signals of the ics , power supply blocks for distribution supply to motors. After finalising the layout of the board and dimensions modelling of the board was carried out. In this phase, modelling of the motor controller of the motor controller board was carried out in Ansys Icepack software package, which is a dedicated software for analysing the thermal efficiency of the circuits. In modelling phase, the components such as Pcb, motor controller, mosfets, resistors, opto coupler and diodes are fabricated. Here in this simulation motor controller Ic is considered as the main heating source.

- In this thesis, the simulation of the modelled circuit was executed in 4 states and the temperature distribution of each stage was evaluated. In the first stage the modelled circuit was simulated in natural convection condition with flow regime of the enclosure module as laminar. The main aim of this stage is to simulate the condition that is widely used in the integrated circuits boards. The main objective of this stage was to reduce the overall temperature value of the modelled circuit. In the second stage heat sink was employed on top of ic having high temperature. Here in this case the main controller ic is the heating element and heat sink was fabricated with reference to the top dimensions of the controller ic and simulated the board under same environment conditions as mentioned in the first stage. In the third stage optimization of the modelled heat sink was carried out with setting parameters like base width of heat sink, number of fins of heat sink, gap between fins of heat sinks .For carrying out optimization procedure ,parametric tool box was linked with ansys icepack software were ansys specifies a range of optimized values and icepack computes the thermal efficiency of board at dimensions of heat sink model and finds dimensions of heat sink were global temperature of circuit is minimum. The simulation environment for the third stage of the circuit is same as that of the first two stages with natural convection and flow regime as laminar. Finally in the fourth stage model was simulated under forced convection condition. A fan was employed in the circuit, whose flow direction is parallel to the direction of the fins. In this stage the flow regime is considered as turbulent and model was simulated with k –epsilon turbulent model.

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- In the first stage of analysis process, after the meshing process, the temperature distribution of the circuit was evaluated without the application of heat sink. Here in the first stage heat flow condition was selected as natural convection with fluid flow type as laminar. The model was iterated for 1000 times and after convergence of the model is met, the computed temperature for motor controller ic was found to 154°C.To reduce the temperature of ic , heatsink was employed in the model with parameters equal to the base geometry of heat sink. The model again was iterated with same environment conditions and the final obtained temperature was found to be 126°C.
- In this thesis, for further reducing the temperature of motor controller ic, optimization procedure was carried out in order to further reduce the temperature from 126°C.For optimization of heatsink, the parameters considered in this simulation are the number of fins, thickness of fin and base width of heat sink. The main objective function of optimization was set to minimize the heat sink temperature. After setting conditions for optimization procedure like upper and lower bound for parameters, optimization routine was executed with 200 iterations for each point. The Final temperature obtained after optimization of heat sink was found to be 108°C with same conditions that was set for previous conditions. After this stage, the model was again evaluated under forced convection condition, with the addition of fan which is set in a direction parallel to the fin axis. In this case the model was evaluated with fluid flow as turbulent using k-epsilon model. After simulating the model, the temperature of 54°C was obtained. Then particle trace, velocity vector, pressure and velocity distribution of the model was finally evaluated.
- In nutshell, in the first stage stage that is condition of natural convection and flow regime as laminar model achieved a maximum global temperature of 154oC. In the second stage, with heat sink on top of controller ic, the model reaches a temperature upto 126oC with similar condition as that of the first stage. In the third stage optimization of the heat sink was carried out and global temperature of model reduced to 108oC under similar conditions as that above 2 stages. In the last stage with the deployment of fan having low power consumption whose flow direction was set parallel to the fins, the model achieved 54oC which shows that with right placement of electronic components, combined with optimization will reduce the need the of additional high power equipment's for cooling.

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- After simulating the modelled geometry for four cases, the study found that thermal efficiency of electronic circuit can achieved very effectively with the help of CFD simulations. In most cases heat sinks are used in circuits for thermal management. In this thesis simulation showed that the optimization of heat sink can very effectively reduce the overall temperature of board. The effective of fan in thermal management is very evident. Moreover, the placement of more heat emitting ics towards the centre of board can greatly improve the thermal efficiency of the board

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